Explicit upper bounds for the Erdős–Surányi function $g(n)$

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Abstract

For $n \geq 1$ let $g(n)$ be the least integer g such that for every set $\{a_1 < \cdots < a_n\}$ of integers greater than 1 and every $x \geq 0$, some g of the integers $x+1, \ldots, x+a_n$ have product divisible by $a_1 \cdots a_n$. Erdős and Surányi (1959) proved $g(3) = 4$ and $g(n) \geq (2 - o(1))n$, and Erdős asked (offering \$100) whether $g(n) \leq (2 + o(1))n$, or even $g(n) \leq 2n$ (Erdős Problem #708). To our knowledge no upper bound depending on n alone has been published. We prove

$$g(n) \;\leq\; n\big(\lceil \log_2 2n \rceil + \lceil \log_2 \lceil \log_2 2n \rceil \rceil + 2\big) \quad (n \geq 2), \qquad g(n) \;\leq\; \left\lceil 48\, n \sqrt{\tfrac{81 + \ln n}{\ln(81 + \ln n)}} \right\rceil,$$

and $g(n) \leq (2 + o(1))\, n \sqrt{\ln n / \ln \ln n}$, and, by a different method (linear-programming duality and an elementary sieve lemma for intervals), $g(n) \leq \big(16 + 20 \ln \ln(8n^3)\big) n = O(n \ln \ln n)$. The duality reduces a linear bound to a "hinge inequality" for weighted prime-factor counts on intervals; we prove it for 0/1 weights with the absolute threshold 4 for every set of primes, by explicit counting certificates; for fractional weights we prove it with threshold 65 whenever the small-modulus part of the weight has mean at least $\frac{17}{16}$, which reduces a linear bound $g(n) \leq 81n$ to a "sparse core" statement; we prove that statement for all $m \leq 10^{2887}$, so that $g(n) \leq 81n$ for all $n \leq 10^{962}$, and we show that the two-sided interval counts alone cannot prove it in general, which remains open. We also show that the conjectured bound holds for long intervals: if $a_n \geq 8n^3$ then $2n$ integers always suffice, so Erdős's question is reduced to the range $a_n < 8n^3$; more generally, for every integer $k \geq 2$, kn integers suffice whenever $a_n \leq kn$ or $a_n^{k-1} \geq (kn)^{k+1}$; hence for every fixed $\delta > 0$ a bound linear in n (with a constant depending on δ) holds whenever $a_n \geq n^{1+\delta}$, e.g. $12n$ integers suffice whenever $a_n \geq n^2$. All proofs are elementary and constructive: prime-power atoms of the a_i that are large relative to a_n are placed one per interval element by a per-prime greedy rule, the small atoms are packed into bins, and a counting argument bounds the number of bins. We also record $g(4) \geq 5$ and $g(5) \geq 6$ by exact computation.

*__Disclosure of the production pipeline.__ Every mathematical statement in this note was produced by automated systems inside an autonomous research pipeline: the proofs of Theorems 1.1 and 1.2 were found in two independent runs of OpenAI GPT-5.6 at "Pro" effort through the web interface (26 and 90 minutes of reasoning respectively, on 2026-09-03) and the proofs of Theorems 1.3 and 1.4 by Qwen3.8-Max (Alibaba) through its web interface in two runs on 2026-09-03/04, all working from briefs written by an orchestrating agent (Claude Fable 5.1, Anthropic) that selected the problem, pinned the statement to Erdős's 1992 text, verified the lower-bound construction by exact computation, re-derived every step of both proofs, commissioned independent referee runs (Claude Opus 5 agents in fresh contexts, which also brute-forced the lemmas on hundreds of thousands of instances), and drafted this document. The four constructions were implemented and exercised on more than seven million random instances, and every numerical inequality in the explicit parameter choices was checked by machine (scripts in the accompanying repository). No human supplied a definition, a lemma, a proof idea, or a correction. The problem itself remains open: the conjectured bound $g(n) \leq 2n$ is not proved here.

1 Introduction

Gallai observed that among any a_2 consecutive integers one can always find two whose product is a multiple of a_1a_2 (for $1 < a_1 < a_2$), but that there are $a_1 < a_2 < a_3$ and a run of a_3 consecutive integers among which no three have product divisible by $a_1a_2a_3$. Erdős and Surányi [1] studied the general situation. Following Erdős's formulation [2, §1], let $g(n)$ be the smallest integer such that, whenever $1 < a_1 < \cdots < a_n$ are integers and $x \geq 0$, there are integers $x < u_1 < \cdots < u_{g(n)} \leq x + a_n$ with $u_1 \cdots u_{g(n)} \equiv 0 \pmod{a_1 \cdots a_n}$. (Since enlarging B inside the interval preserves divisibility, "$g(n)$ integers" may be read as "at most $g(n)$ integers"; we use the latter, which also covers the case $a_n < g(n)$.) They proved $g(3) = 4$ and

$$g(n) \;\geq\; (2-\varepsilon)\, n \qquad (n > n_0(\varepsilon)), \tag{1}$$

the lower bound coming from the set of all products p_ip_j of ℓ primes satisfying $2p_1^2 > p_\ell^2$, with the interval positioned so that a single integer u, divisible exactly once by each p_i, is the only common multiple of every p_ip_j in it. Erdős [2] asked whether $g(n) < (2+\varepsilon)n$ for large n, "or perhaps even $g(n) \leq 2n$", and offered \$100 for a proof or disproof; the question is Problem #708 on the Erdős problems site [3]. Neither [2] nor [3] records an upper bound for $g(n)$, and we are not aware of one in the literature (the original paper [1], in Hungarian, was not accessible to us). Note that a bound in terms of n alone is not obvious: the trivial estimate $g \leq \sum_i \omega(a_i)$ (one interval element per prime factor of each a_i, see Lemma 2.3) depends on a_n, which is unbounded.

Throughout, $\mathrm{v}_p(k)$ is the exponent of the prime p in k, $\omega(k)$ the number of distinct prime factors of k, and $\Lambda(t) = \lceil \log_2 t \rceil$ is the least $r \geq 0$ with $t \leq 2^r$.

Theorem 1.1. *For every $n \geq 2$,*

$$g(n) \;\leq\; F(n) := n\big(\Lambda(2n) + \Lambda(\Lambda(2n)) + 2\big) \;=\; n\log_2 n + n\log_2\log_2 n + O(n).$$

In particular $F(2) = 10$, $F(3) = 21$, $F(4) = 28$, $F(10) = 100$, $F(100) = 1300$, $F(1000) = 17000$.

Theorem 1.2. *For every $n \geq 1$, with $L = 81 + \ln n$,*

$$g(n) \;\leq\; \left\lceil 48\, n\sqrt{L/\ln L}\,\right\rceil, \qquad \textit{and} \qquad g(n) \;\leq\; (2+o(1))\, n\sqrt{\frac{\ln n}{\ln\ln n}} \quad (n \to \infty).$$

Theorem 1.1 gives the smaller value for every n below about 10^{104}; Theorem 1.2 is asymptotically stronger. Neither is close to the conjectured $2n$. For long intervals, however, the conjecture holds:

Theorem 1.3. *If $a_n \geq 8n^3$, then for every $x \geq 0$ some $2n$ of the integers $x+1, \ldots, x+a_n$ have product divisible by $a_1 \cdots a_n$ (at most $2n$ are needed, and since $a_n \geq 2n$ the set may be padded); in fact Conjecture 7.4 below holds for such A. Consequently $g(n) \leq 8n^3$ as well, and the question whether $g(n) \leq 2n$ is equivalent to the same question restricted to $a_n < 8n^3$.*

Theorem 1.3 is the case $k = 2$ of a one-parameter family:

Theorem 1.4. *Let $k \geq 2$ be an integer and suppose that $a_n \leq kn$ or $a_n^{k-1} \geq (kn)^{k+1}$. Then for every $x \geq 0$ some (at most) kn of the integers $x+1, \ldots, x+a_n$ have product divisible by $a_1 \cdots a_n$. In particular $a_n \geq 9n^2$ gives $3n$ and $a_n \geq 1024^{1/3}n^{5/3}$ gives $4n$; for every $\delta > 0$ and every integer k with $\delta(k-1) > 2$, $a_n \geq n^{1+\delta}$ gives kn as soon as $n^{\delta(k-1)-2} \geq k^{k+1}$. Taking for each n the smallest admissible k (and Theorem 1.1 where none is admissible) gives, for every $n \geq 1$: $7n$ integers suffice whenever $a_n \geq n^3$, $12n$ whenever $a_n \geq n^2$, and $15n$ whenever $a_n \geq n^{7/4}$.*

Thus for every fixed $\delta > 0$ a bound linear in n holds when $a_n \geq n^{1+\delta}$, with a constant depending on δ; for a fixed constant C the range not covered by Theorem 1.4 is $Cn < a_n < (Cn)^{(C+1)/(C-1)}$, a full power of n. The Erdős–Surányi family, with $a_n \asymp n(\ln n)^2$, lies just above the trivial range. The gap between $n\sqrt{\ln n/\ln\ln n}$ and $2n$ is discussed in Section 7.

Finally, Section 8 reduces a linear bound to a single inequality about additive functions on intervals (Conjecture 8.1): if it holds, then $g(n) \leq 18n$ (Theorem 8.2). We prove the inequality in two special cases and show, with explicit certificates, that three natural strengthenings of it are false because intervals can contain more integers free of small primes than $[1, m]$ does. Section 9 shows that the inequality follows, by linear-programming duality, from the two-sided count $\lfloor m/d \rfloor \leq N_I(d) \leq \lfloor m/d \rfloor + 1$ of multiples alone in every instance we could compute, observes that for the full set of primes the 0/1 case already follows from the trivial bound $\sum_I (w-1)^+ \geq \sum_{k \leq m} w(k) - m$ (Corollary 9.2), gives an explicit anchor certificate for prime sets without small primes, where that bound fails, and proves the 0/1 case for all m and all prime sets with the threshold 2 replaced by $20 \ln\ln m$ (Theorem 9.4); Section 12 replaces $20 \ln\ln m$ by the absolute threshold 4, for every prime set, with two explicit certificates (Theorem 12.1). Section 10 extends this to arbitrary weights with threshold $1 + 149e \ln(1 + \ln m)$ (Theorem 10.1), which through the same duality gives $g(n) = O(n \ln\ln n)$ (Corollary 10.2). Section 11 sharpens the threshold to $20 \ln\ln m$ by a geometric refinement of the same sieve (Theorem 11.1), giving $g(n) \leq (16 + 20 \ln\ln(8n^3))n$ for all n, which is below the bound of Theorem 1.2 for every n. Section 13 returns to the fractional case: the inequality with threshold 65 holds whenever the small-modulus atoms of the weight have mean at least $\frac{17}{16}$ (Theorem 13.2), so a linear bound $g(n) \leq 81n$ follows from a "sparse core" statement (Corollary 13.3); we prove that statement for $m \leq 10^{2887}$ by an arithmetic argument (Theorem 13.7), which gives $g(n) \leq 81n$ for all $n \leq 10^{962}$ (Corollary 13.8), and we show that the two-sided interval counts alone cannot prove it in general (Proposition 13.5).

2 The common construction

Fix $A = \{a_1 < \cdots < a_n\} \subset \{2, 3, \ldots\}$, put $m = a_n$, $I = \{x+1, \ldots, x+m\}$ and $P = a_1 \cdots a_n$. We must find $B \subseteq I$ with $P \mid \prod_{b \in B} b$ and bound $|B|$. Note that B is a set: an interval element contributes its valuations once, however many a_i it may be a multiple of.

Lemma 2.1 (counting). *For every integer $d \leq m$: $\#\{a \in A : d \mid a\} \leq \lfloor m/d \rfloor \leq \lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor = \#\{b \in I : d \mid b\}$.*

Proof. The multiples of d in $\{1, \ldots, m\}$ are $d, 2d, \ldots, \lfloor m/d \rfloor d$, and the a_i are distinct elements of $\{2, \ldots, m\}$. Any m consecutive integers contain $\lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor \geq \lfloor m/d \rfloor$ multiples of d. □

Lemma 2.2 (whole interval). *P divides $\prod_{b \in I} b$. Hence if $m \leq H$ then $B = I$ has $|B| = m \leq H$.*

Proof. $P \mid m!$ because the a_i are distinct elements of $\{2, \ldots, m\}$, and $\prod_{b \in I} b = m!\binom{x+m}{m}$. □

Atoms, large and small. Fix an integer budget $H \geq 1$ and assume $m > H$; put $T = m/H > 1$. For $a \in A$ and $p \mid a$ call $q_p(a) = p^{\mathrm{v}_p(a)}$ an *atom* of a; the atoms of a are pairwise coprime and multiply to a, and a has $\omega(a)$ atoms. An atom is *large* if $q > T$ and *small* if $q \leq T$.

Lemma 2.3 (large atoms, one element each)**.** *Let $\mathcal{L}$ be the multiset of all large atoms of all $a \in A$ (atoms of different a are distinct members, so $r = |\mathcal{L}|$ counts with multiplicity). There is a* set *$U \subseteq I$ with $|U| \le r$ such that for every prime p*

$$\mathrm{v}_p\Big(\prod_{u\in U} u\Big) \;\ge \sum_{q\in\mathcal{L},\; q \text{ a power of } p} \mathrm{v}_p(q).$$

More precisely, one can assign to each $q \in \mathcal{L}$ an element $\beta(q) \in I$ with $q \mid \beta(q)$ such that atoms of the same prime receive distinct elements; then $U = \beta(\mathcal{L})$.

Proof. Fix a prime p and list the large atoms of p as $p^{e_1}, \dots, p^{e_s}$ with $e_1 \ge \cdots \ge e_s$. Each $a \in A$ has at most one atom of p, so the first j atoms come from j distinct elements of A, all divisible by p^{e_j}; by Lemma 2.1, I contains at least j multiples of p^{e_j}. Assign the atoms in this order, giving the j-th one a multiple of p^{e_j} in I different from the $j-1$ elements already used for p. Elements used for different primes may coincide; this only decreases $|U|$. For a fixed p the assigned elements are distinct, so their p-adic valuations add up to at least $\sum_j e_j$. □

Lemma 2.4 (assembly)**.** *Suppose the small atoms of all $a \in A$ (a multiset; atoms of different a are distinct members) have been partitioned into* bins*, each bin being a set of small atoms whose product f satisfies $f \le T$ (a bin may contain atoms of different a). Let s be the total number of bins and $D = r + s$. If $D \le H$, then there is $B \subseteq I$ with $|B| \le D$ and $P \mid \prod_{b\in B} b$.*

Proof. Take U from Lemma 2.3, $|U| \le r$. Order the bins $f_1, \dots, f_s$ arbitrarily. Since $f_j \le T = m/H$ and H is an integer, Lemma 2.1 gives at least $\lfloor m/f_j \rfloor \ge H$ multiples of f_j in I. When bin j is processed, the elements to avoid are those of U and the $j-1$ elements chosen for earlier bins, at most $r + s - 1 = D - 1 \le H - 1$ in number; so an unused multiple v_j of f_j exists. Put $B = U \cup \{v_1, \dots, v_s\}$. By construction $U \cap \{v_1, \dots, v_s\} = \emptyset$ and the v_j are pairwise distinct, so $|B| \le r + s = D$ and (this is the only place where the valuations of different elements are added)

$$\prod_{b\in B} b = \prod_{u\in U} u \cdot \prod_j v_j,$$

whose p-adic valuation is at least (sum over large atoms of p) $+\ \mathrm{v}_p(\prod_j f_j)$, which equals $\mathrm{v}_p(P)$ because every atom of every a is either large or lies in exactly one bin. □

Thus everything reduces to producing bins and a budget H with $D = r+s \le H$. Theorems 1.1 and 1.2 do this in two different ways.

3 Proof of Theorem 1.1

Next-fit bins. For each $a \in A$ list its small atoms in increasing order of the prime and pack them by multiplicative next-fit: keep a current bin, initially empty (product 1); if multiplying the next atom keeps the product $\le T$, add it; otherwise close the bin and start a new one containing that atom. Every small atom is $\le T$, so every bin has product $\le T$.

Lemma 3.1. *If $d_1, \dots, d_u$ are the bin products of one a in the order created, then $d_j d_{j+1} > T$ for $1 \le j < u$.*

Proof. Bin j was closed because the next atom q did not fit: $d_j q > T$. That atom is the first element of bin $j+1$, so $d_{j+1} \ge q$. □

Lemma 3.2 (budget)**.** *Let $H, \kappa \geq 1$ be integers with*

$$\text{(B1)}\ \ H \geq n(2\kappa+1), \qquad \text{(B2)}\ \ H^{(\kappa+1)n} \leq 2^{\kappa H}. \tag{2}$$

Then the next-fit construction with threshold $T = m/H$ produces $D = r + s \leq H$ demands.

Proof. *Case $m^\kappa \leq H^{\kappa+1}$.* Binning never increases the count, so $D \leq W := \sum_{a\in A}\omega(a)$. Since every atom is at least 2, $2^{\omega(a)} \leq a$, hence $2^W \leq \prod_a a \leq m^n$ and $2^{\kappa W} \leq m^{\kappa n} \leq H^{(\kappa+1)n} \leq 2^{\kappa H}$ by (B2). So $W \leq H$.

Case $m^\kappa > H^{\kappa+1}$. Then $T^{\kappa+1} = m^{\kappa+1}/H^{\kappa+1} > m$. Fix $a \in A$ with ℓ large atoms and u bins $d_1, \ldots, d_u$. The large atoms and the products d_1d_2, $d_3d_4, \ldots$ of consecutive bin pairs are pairwise coprime divisors of a, each exceeding T (Lemma 3.1). If $\ell + \lfloor u/2 \rfloor \geq \kappa+1$, the product of $\kappa+1$ of them would exceed $T^{\kappa+1} > m \geq a$ while dividing a; so $\ell + \lfloor u/2\rfloor \leq \kappa$, and since $u \leq 2\lfloor u/2 \rfloor + 1$, $\ell + u \leq 2(\ell + \lfloor u/2 \rfloor) + 1 \leq 2\kappa+1$. Summing over a, $D \leq n(2\kappa+1) \leq H$ by (B1). ☐

Lemma 3.3. *For $n \geq 2$ let $r = \Lambda(2n)$, $s = \Lambda(r)$, $C = r + s + 2$, $H = nC = F(n)$ and $\kappa = \lfloor (C-1)/2 \rfloor$. Then* (B1) *and* (B2) *hold.*

Proof. (B1): $2\kappa + 1 \leq C$. (B2), $n \geq 3$: then $2n \geq 6$, so $r \geq 3$ and $s \geq 2$; by definition $2n \leq 2^r$, i.e. $n \leq 2^{r-1}$, and $r \leq 2^s$; also $s + 2 \leq 2^s$ for $s \geq 2$. Hence $C = r + s + 2 \leq 2^{s+1}$ and $H = nC \leq 2^{r-1}2^{s+1} = 2^{C-2}$. Moreover $C \leq 2(\kappa+1)$, i.e. $(C-2)(\kappa+1) \leq \kappa C$, so $H^{(\kappa+1)n} \leq 2^{(C-2)(\kappa+1)n} \leq 2^{\kappa C n} = 2^{\kappa H}$. For $n = 2$: $r = 2$, $s = 1$, $C = 5$, $H = 10$, $\kappa = 2$, and $10^6 \leq 2^{20}$. ☐

Proof of Theorem 1.1. Let $H = F(n)$. If $m \leq H$ use Lemma 2.2. Otherwise Lemmas 3.2 and 3.3 give $D \leq H$, and Lemma 2.4 gives B with $|B| \leq D \leq F(n)$. ☐

4 Proof of Theorem 1.2

Here the bins are formed by *maximal merging*: start with each small atom (of any a) in its own bin and, as long as two bins have product $\leq T$, merge them. Each merge preserves the property "product $\leq T$" and decreases the number of bins, so the process terminates, and at termination any two bins have product $> T$.

Lemma 4.1. *With $W = \sum_a \omega(a)$ and $D = r + s$ as before,*

$$\text{(i)}\ \ D \leq W, \qquad \text{(ii)}\ \ D \ \leq\ \frac{2n\ln m}{\ln(m/H)} + 1.$$

Proof. (i) is clear. (ii): any two bins have product $> T$, so at most one bin has product $\leq \sqrt{T}$, and that bin still has product $\geq 2 > 1$. Hence, if $s \geq 1$, $P = \prod_{\text{large}} q \cdot \prod_j f_j > T^r\, T^{(s-1)/2}$ strictly, and $P \leq m^n$; thus $r + (s-1)/2 < n\ln m/\ln T$ and $D = r + s < 2n\ln m/\ln T - r + 1 \leq 2n\ln m/\ln(m/H) + 1$. If $s = 0$, $P > T^r$ gives $D = r < n\ln m/\ln T$. ☐

Lemma 4.2. *For real $X > 1$, $\sum_{p\leq X} 1/p \leq e\ln(1+\ln X)$ (sum over primes).*

Proof. Put $\sigma = 1 + 1/\ln X$. For $p \leq X$, $p^{\sigma-1} \leq X^{1/\ln X} = e$, so $1/p \leq e\,p^{-\sigma}$. Using $-\ln(1-t) \geq t$ and the finite Euler product, $\sum_{p\leq X} p^{-\sigma} \leq \ln\prod_{p\leq X}(1-p^{-\sigma})^{-1} \leq \ln\zeta(\sigma)$, and $\zeta(\sigma) \leq 1 + \int_1^\infty t^{-\sigma}dt = 1 + 1/(\sigma-1) = 1 + \ln X$. ☐

Lemma 4.3 (average number of prime factors)**.** *For every real $t > 1$,*

$$\frac{1}{n}\sum_{i=1}^{n}\omega(a_i) \;\le\; \frac{\ln(m/n) + (t-1)\,e\ln(1+\ln m)}{\ln t}.$$

Proof. For every positive integer k, $t^{\omega(k)} = \prod_{p\mid k}\bigl(1+(t-1)\bigr) = \sum_{d\mid k,\ d \text{ squarefree}}(t-1)^{\omega(d)}$. All terms are nonnegative because $t > 1$. Summing over $k \le m$,

$$\sum_{k\le m} t^{\omega(k)} = \sum_{\substack{d\le m\\ d \text{ sqfree}}} (t-1)^{\omega(d)}\Bigl\lfloor\frac{m}{d}\Bigr\rfloor \le m\prod_{p\le m}\Bigl(1+\frac{t-1}{p}\Bigr) \le m\exp\Bigl((t-1)\sum_{p\le m}\frac{1}{p}\Bigr) \le m\,e^{(t-1)e\ln(1+\ln m)}$$

by Lemma 4.2. The a_i are distinct integers in $[2,m]$, so $\sum_i t^{\omega(a_i)} \le \sum_{k\le m} t^{\omega(k)}$, and by the AM–GM inequality $t^{\frac{1}{n}\sum_i \omega(a_i)} \le \frac{1}{n}\sum_i t^{\omega(a_i)}$. Taking logarithms gives the claim (note $m \ge n+1$, so $\ln(m/n) > 0$). □

Lemma 4.4 (explicit parameters)**.** *Let $L = 81 + \ln n$, $\ell = \ln L$, $R = \sqrt{L/\ell}$ and $H = \lceil 48nR\rceil$. If $m > H$ then $D < H$.*

Proof. Elementary facts: $\ell > 4$ (as $L \ge 81 > e^4$); $\ell \le L/4$ (the function $y/4 - \ln y$ is increasing and positive for $y \ge 81$); $\ln\ell \le \ell/2$ (as $y/2 - \ln y > 0$ is increasing for $y \ge 4$); hence $R > 2$. Put $y = \ln m$, $z = \ln H$, $d = y - z > 0$, $c = z - \ln n$. Since $48nR > 1$, $H \le 96nR$, so $c \le \ln 96 + \ln R < 7 + \ell/2$ (using $\ln R = (\ell - \ln\ell)/2 \le \ell/2$) and $z < \ln n + 7 + \ell/2 < 2L$. Put $\Delta = 4\sqrt{L\ell} = 4R\ell \le 2L$ (as $4\ell \le L$).

Case $d \le \Delta$. Then $y \le 4L$. Apply Lemma 4.3 with $t = R$: $\ln R = (\ell - \ln\ell)/2 \ge \ell/4$ and $e\ln(1+y) \le e\ln(5L) < 3(\ell+3) < 6\ell$. Since $\ln(m/n) = d + c$,

$$\frac{W}{n} \le \frac{d + c + (R-1)\cdot 6\ell}{\ell/4} < \frac{4R\ell + 7 + \ell/2 + 6R\ell}{\ell/4} = 40R + \frac{28}{\ell} + 2 < 45R,$$

so by Lemma 4.1(i), $D \le W < 45nR < H$.

Case $d > \Delta$. By Lemma 4.1(ii), $D \le 2n\,\frac{z+d}{d} + 1 < 2n + \frac{2n\cdot 2L}{4\sqrt{L\ell}} + 1 = 2n + nR + 1 < 3nR < H$, using $R > 2$. □

Proof of Theorem 1.2, explicit part. Let $H = \lceil 48nR\rceil$. If $m \le H$ use Lemma 2.2; otherwise Lemma 4.4 gives $D < H$ and Lemma 2.4 gives $|B| \le D < H$. □

Proof of Theorem 1.2, asymptotic part. Fix $\varepsilon > 0$, put $Q_n = \sqrt{\ln n/\ln\ln n}$ and $H = \lceil (2+\varepsilon)nQ_n\rceil$; we show $D < H$ for all $m > H$ once n is large, which suffices as before. With $y = \ln m$, $z = \ln H$, $d = y - z$, $c = z - \ln n$ we have $z = \ln n + O(\ln\ln n)$, hence $\sqrt{z/\ln z} = (1+o(1))Q_n$ and $c = O(\ln z)$. Put $\Delta = \sqrt{z\ln z}$. If $d > \Delta$, Lemma 4.1(ii) gives $D/n \le 2 + 2z/d + 1/n \le (2+o(1))\sqrt{z/\ln z}$. If $d \le \Delta$, then $y = (1+o(1))z$; apply Lemma 4.3 with $t = \sqrt{z}/(\ln z)^2$, so $\ln t = (\frac{1}{2}+o(1))\ln z$: the three terms of the numerator divided by $\ln t$ are at most $(2+o(1))\sqrt{z/\ln z}$, $O(1)$ and $O(\sqrt{z}/(\ln z)^2)$ respectively, so $D/n \le W/n \le (2+o(1))\sqrt{z/\ln z}$. In both cases $D \le (2+\varepsilon/2)nQ_n < H$ for all $n \ge n_0(\varepsilon)$, so $g(n) \le \lceil (2+\varepsilon)nQ_n\rceil$ for $n \ge n_0(\varepsilon)$; as $\varepsilon > 0$ was arbitrary, $\limsup_n g(n)/(nQ_n) \le 2$. □

Remark 4.5. No attempt was made to optimise the constant 48; the same argument with the case threshold and the prime-sum estimate tightened (e.g. an explicit Mertens bound in place of Lemma 4.2) gives a single-digit constant. Theorems 1.1 and 1.2 are proved by the same construction and differ only in how the number of bins is counted; Theorem 1.3 uses the same two ingredients (per-prime placement of large atoms, private multiples for small parts) with the budget $2n$ made possible by the length of the interval.

5 Long intervals: proof of Theorem 1.3

Assume $m = a_n \geq 8n^3$ and put $C = m/(2n)$. Then $C \geq m^{2/3}$ (equivalent to $m^{1/3} \geq 2n$), $C \geq 4n^2 \geq 2n$, and, since $m \geq 8n^3 \geq 4n^2$, also $C^2 = m^2/(4n^2) \geq m$. In the language of Section 2 we take the integer budget $H = 2n$ and threshold $T = C = m/H$ (note $m > H$).

Lemma 5.1 (two-bin packing)**.** *Let $C \geq 1$ be real and let $q_1, \ldots, q_r > 1$ be integers with $q_i \leq C$ and $P = q_1 \cdots q_r \leq C^{3/2}$. Then some sub-product s of the q_i (over a subset, possibly empty or everything) satisfies $P/C \leq s \leq C$; consequently the q_i can be partitioned into two classes with products $\leq C$.*

Proof. If $P \leq C$ take $s = P$. Otherwise $1 < P/C \leq \sqrt{C}$. If some $q_i > \sqrt{C}$, take $s = q_i$: then $s \leq C$ and $s > \sqrt{C} \geq P/C$. If all $q_i \leq \sqrt{C}$, multiply them in any order and stop at the first partial product $s \geq P/C$ (this happens at the latest at $s = P$); the previous partial product, possibly the empty product $1 < P/C$, was $< P/C$, so $s < (P/C)\sqrt{C} = P/\sqrt{C} \leq C$. In every case the complementary product is $P/s \leq C$. $\square$

Lemma 5.2 (split)**.** *Every $a \in A$ has at most one atom exceeding C, and admits a factorisation $a = d_1 d_2$ into coprime parts such that either $d_1, d_2 \leq C$, or d_1 is an atom $> C$ and $d_2 < 2n \leq C$.*

Proof. Two atoms $> C$ are coprime, so they would give $a > C^2 \geq m$, impossible. If $q = p^e > C$ is an atom, put $d_1 = q$, $d_2 = a/q < m/C = 2n$. Otherwise all atoms of a are $\leq C$ and their product $a \leq m \leq C^{3/2}$, so Lemma 5.1 splits them into two coprime classes with products $\leq C$. $\square$

Proof of Theorem 1.3. Split every $a \in A$ by Lemma 5.2; parts equal to 1 impose no condition and may be assigned to any element, so we discard them. This gives at most $2n$ *demands*, each either a *large* atom $> C$ or a *small* integer $\leq C$. (Lemma 2.4 with $H = 2n$ already gives $|B| \leq 2n$; we repeat the argument because Conjecture 7.4 asks for the explicit assignment.) Assign the large demands as in Lemma 2.3: for each prime p, the large atoms of p (one per a) receive distinct multiples of the required powers of p; different primes may share elements. Let U be the set of elements used, $|U| \leq$ number of large demands. Now assign the small demands one at a time: a small demand $d \leq C = m/(2n)$ has at least $\lfloor m/d \rfloor \geq 2n$ multiples in I by Lemma 2.1 ($2n$ is an integer), while the forbidden elements (U together with the elements already given to earlier small demands) number at most $2n - 1$, because there are at most $2n$ demands in all and the current one is not yet placed; so d receives a multiple of its own, outside U and different from all other small representatives. Let B be the set of all elements used, $|B| \leq 2n$. At an element of U the demands present are prime powers of distinct primes, each dividing the element; at every other element of B exactly one demand d is present and d divides the element. Hence the product of the demands at any $b \in B$ divides b, and since the demands of each a multiply to a, $P \mid \prod_{b \in B} b$. This is exactly an assignment as in Conjecture 7.4. Finally, if $a_n < 8n^3$ then Lemma 2.2 gives a solution with at most $a_n < 8n^3$ elements, so $g(n) \leq 8n^3$. $\square$

6 The k-split lemma: proof of Theorem 1.4

Lemma 6.1 (k-split)**.** *Let $k \geq 1$ and let $H \leq m$ be positive integers. Let Q be a finite multiset of integers > 1 with product A such that $A^2H^{k+1} \leq m^{k+1}$. Then Q can be partitioned into at most k pieces (none if $Q = \emptyset$), each of which is either a single element of Q or a* bin*: a nonempty $F \subseteq Q$ whose product f satisfies $fH \leq m$.*

Proof. Induction on k; if $Q = \emptyset$ there is nothing to do, so let $Q \neq \emptyset$. For $k = 1$ the hypothesis reads $A^2H^2 \leq m^2$, so $AH \leq m$ and Q itself is one bin. Let $k \geq 2$.

Case 1: some $q \in Q$ has $q^2H > m$. Make q a singleton piece. The product $A' = A/q$ of the rest satisfies $A'^2H^k = A^2H^{k+1}/(q^2H) < m^{k+1}/m = m^k$, so the induction hypothesis (with $k-1$) splits the rest into at most $k-1$ pieces.

Case 2: $q^2H \leq m$ for all $q \in Q$. Build one bin greedily: start with $f = 1$ and, while Q is not exhausted and $f^2H < m$, multiply the next element into f. The invariant $fH \leq m$ holds at the start ($H \leq m$) and is preserved, because before a step $f^2H < m$ and $q^2H \leq m$ give $(fqH)^2 < m^2$. The first element is always absorbed unless $H = m$, in which case the hypothesis forces $A = 1$, i.e. $Q = \emptyset$; so the bin is nonempty. If the process exhausts Q, all of Q is one bin. Otherwise it stops with $f^2H \geq m$; then the product $A' = A/f$ of the remaining elements satisfies $A'^2H^k = A^2H^{k+1}/(f^2H) \leq m^k$, and the induction hypothesis splits them into at most $k-1$ further pieces. □

Proof of Theorem 1.4. Put $H = kn$. If $m \leq H$, Lemma 2.2 gives $B = I$ with $|B| = m \leq kn$. Otherwise $m > H$ and $m^{k-1} \geq H^{k+1}$; for every $a \in A$, $a^2H^{k+1} \leq m^2 \cdot m^{k-1} = m^{k+1}$. Apply Lemma 6.1 to the atoms of each $a \in A$ (pairwise coprime, product a, all $\leq m$): each a splits into at most k pieces, hence at most $kn = H$ pieces in all. Call a singleton piece q *large* if $qH > m$; every other piece (a bin, or a singleton with $qH \leq m$) is a bin f with $fH \leq m$ in the sense of Section 2 with threshold $T = m/H$. The large pieces are atoms of distinct elements of A for each fixed prime (an element of A has one atom per prime), so Lemma 2.3 applies to them, and Lemma 2.4 (with r = number of large pieces, s = number of bins, $r + s \leq H$) gives $B \subseteq I$ with $|B| \leq H = kn$ and $P \mid \prod_{b \in B} b$; note that the p-adic valuation of a bin comes from the single p-atom of the element of A it was cut from. For the corollaries: $k = 2$ is $m \geq 8n^3$; $k = 3$ is $m^2 \geq 81n^4$; $k = 4$ is $m^3 \geq 1024n^5$; for $\delta > 0$ and $\delta(k-1) > 2$, $m \geq n^{1+\delta}$ gives $m^{k-1} \geq n^{(1+\delta)(k-1)} \geq k^{k+1}n^{k+1}$ as soon as $n^{\delta(k-1)-2} \geq k^{k+1}$. The last sentence of the theorem is a finite computation: for each exponent e and each n take the least $k \geq 2$ with $e(k-1)\ln n \geq (k+1)\ln(kn)$, or $F(n)/n$ from Theorem 1.1 if there is none; for $e = 3, 2, 7/4$ some $k \leq 10$ is admissible for all $n \geq 5$, $n \geq 38$, $n \geq 207$ respectively, and the maximum over n is attained at $n = 4, 36, 165$ with the values $7, 12, 15$; the script is in the repository. □

Remark 6.2. Lemma 6.1 is the sharp form of the same atom-binning that underlies Theorem 1.1 (which needs no hypothesis on a_n); what is new here is the exact threshold $\sqrt{m/H}$ and the explicit family of thresholds $a_n^{k-1} \geq (kn)^{k+1}$. The threshold is essentially sharp for the lemma: $k+1$ atoms each with $q^2H > m$ cannot share bins and violate the hypothesis. With $k = 2$ the lemma is Lemma 5.1 in a different normalisation, so Section 5 is the case $k = 2$; it was kept because it also exhibits the explicit assignment required by Conjecture 7.4. Choosing k as a function of m/n does not improve the global bounds of Theorems 1.1 and 1.2: for $m = n\, e^{\sqrt{\ln n \ln\ln n}}$ both routes give about $n\sqrt{\ln n/\ln\ln n}$.

7 Small values, obstructions and remarks

Proposition 7.1. *$g(4) \geq 5$ and $g(5) \geq 6$.*

Proof. Let $A = \{10, 12, 15, 20\}$ and $x = 50$, so $I = \{51, \dots, 70\}$ and $P = 2^5 \cdot 3^2 \cdot 5^3$. No element of I is divisible by 25, so a solution contains at least three of the multiples $55, 60, 65, 70$ of 5; any set S of multiples of 5 in I has $\mathrm{v}_2(\prod S) \leq 3$ and $\mathrm{v}_3(\prod S) \leq 1$. Hence a further element with $\mathrm{v}_2 \geq 2$ and $\mathrm{v}_3 \geq 1$, i.e. a multiple of 12, is needed; the only multiple of 12 in I is 60, and using it still leaves $\mathrm{v}_2 \leq 3 < 5$. So at least five elements are needed, and $\{55, 63, 64, 65, 70\}$ works. For $A = \{5, 10, 12, 15, 20\}$ and $x = 50$ we have $P = 2^5 \cdot 3^2 \cdot 5^4$, so all four multiples of 5 are forced, supplying $\mathrm{v}_2 = 3$ and $\mathrm{v}_3 = 1$; two more elements are then needed for 2^2 and 3 (no element of I other than 60 is divisible by 12), and $\{55, 60, 63, 64, 65, 70\}$ works. Both minima were also confirmed by an exhaustive dynamic programme over capped valuation vectors, which further shows that no other x gives a larger minimum for these two sets. □

Remark 7.2 (why the obvious approaches fail)**.** (a) Choosing a multiple of each a_i and multiplying does not work, because the chosen multiples may coincide and a shared element contributes its valuations once; in the Erdős–Surányi family every a_i has the same unique multiple u. (b) One cannot in general match the $a_i \leq m/2$ to *distinct* multiples inside an interval of length m: van Doorn, Li and Tang [4] showed that an interval of length $2 \max A$ guarantees only $\min(n, \lceil 2\sqrt{n} \rceil)$ disjoint pairs (a, multiple of a), and this is optimal. (c) An induction that removes a_n, takes a solution B' for the rest and repairs it with two new elements fails: for $A' = \{2, 4, 8, 10, 12\}$, $I = \{1, \dots, 16\}$, $B' = \{8, 12, 15, 16\}$ and $a_6 = 16$, the missing factor 2^4 cannot be supplied by two elements of $I \setminus B'$. Lemma 2.4 avoids all three pitfalls by giving every bin at least H candidates and letting different primes share an element only through the per-prime bookkeeping of Lemma 2.3.

Remark 7.3 (the gap to $2n$)**.** Both proofs pay one interval element per large atom and per bin. In the balancing region of Lemma 4.1 an a_i may have about $\sqrt{\ln n / \ln \ln n}$ atoms of "medium" size, none of which can be merged, and this is exactly the loss in Theorem 1.2. A linear bound would require medium atoms of *different* a_i to share representatives. A natural sufficient statement is the following.

Conjecture 7.4 (split assignment)**.** For every A and I as above one can choose for each $a \in A$ a factorisation $a = d_1(a)d_2(a)$ into coprime parts ($d_2 = 1$ allowed) and assign each of the at most $2n$ parts to an element $\beta(d) \in I$ so that for every $b \in I$ and every prime p, $\sum_{d:\beta(d)=b} \mathrm{v}_p(d) \leq \mathrm{v}_p(b)$.

The conjecture implies $g(n) \leq 2n$ (take $B = \beta(\text{parts})$), and Theorem 1.3 proves it whenever $a_n \geq 8n^3$. Since the capacity constraint couples two parts only if they share a prime, the assignment always exists when every $a \in A$ has at most two distinct prime factors (use the finest splitting and Lemma 2.3 prime by prime); a counterexample would need elements with three or more prime factors competing for composite multiples. An exact search found the assignment on the Erdős–Surányi instances with $\ell = 3, 4$, on the instances of Proposition 7.1, and on more than 10^6 random instances with $n \leq 6$ and $a_n \leq 210$; we found no counterexample.

8 A conditional linear bound

Theorem 1.3 settles the range $a_n \geq 8n^3$ with the conjectured value $2n$, so a bound $g(n) \leq Cn$ with an absolute constant C is a statement about $a_n < 8n^3$ only. In this section we reduce such a bound to a single inequality about additive functions on intervals of integers, we prove the inequality in two special cases, and we record three natural strengthenings of it that are *false*, with explicit certificates.

Throughout, a *weight* is a family $z = (z_p)_p$ indexed by the primes with $0 \le z_p \le 1$ and only finitely many $z_p \ne 0$, and $w_z(k) := \sum_p z_p \, \mathrm{v}_p(k)$ is the associated completely additive function. We write $t^+ = \max(t, 0)$.

Conjecture 8.1 (hinge inequality)**.** For every $m \ge 1$, every $x \ge 0$ and every weight z, with $I = \{x+1, \dots, x+m\}$,

$$\sum_{k=1}^{m} \big(w_z(k) - 2\big)^+ \;\le\; \sum_{b \in I} \big(w_z(b) - 1\big)^+. \tag{3}$$

Theorem 8.2. *If Conjecture 8.1 holds, then $g(n) \le 18n$ for all $n \ge 1$. More precisely, it suffices that* (3) *holds for $m = a_n$ and every weight supported on the primes dividing $\prod A$.*

The proof is a linear-programming argument. Fix $A = \{a_1 < \cdots < a_n\}$ with $m = a_n < 8n^3$, an interval I of length m, the set $\mathcal{P}$ of primes dividing $P = \prod A$, and $R_p := \sum_{a \in A} \mathrm{v}_p(a)$ for $p \in \mathcal{P}$. Consider the fractional cover problem

$$\tau^* := \min\Big\{ \sum_{b \in I} y_b : \; \sum_{b \in I} \mathrm{v}_p(b)\, y_b \ge R_p \ (p \in \mathcal{P}), \ 0 \le y_b \le 1 \Big\}. \tag{4}$$

It is feasible, since $y \equiv 1$ works: $\prod A$ divides $m!$, which divides the product of any m consecutive integers, so $\sum_{b \in I} \mathrm{v}_p(b) \ge R_p$.

Lemma 8.3 (duality)**.** $\tau^* = \max_{z \ge 0} \Big[\sum_{a \in A} w_z(a) - \sum_{b \in I} \big(w_z(b) - 1\big)^+ \Big]$, *where the maximum is over $z = (z_p)_{p \in \mathcal{P}}$ with $z_p \ge 0$, and it is attained at some z with all $z_p \le 1$.*

Proof. The dual of (4) has variables $z_p \ge 0$ (one per covering constraint) and $u_b \ge 0$ (one per upper bound $y_b \le 1$) and reads $\max \sum_p z_p R_p - \sum_b u_b$ subject to $\sum_p z_p \, \mathrm{v}_p(b) - u_b \le 1$ for every $b \in I$. For fixed z the best choice is $u_b = (w_z(b) - 1)^+$, and $\sum_p z_p R_p = \sum_{a \in A} w_z(a)$; strong duality holds because (4) is feasible and bounded. For the last claim fix all z_q, $q \ne p$, and let $z_p \ge 1$ vary: every b with $p \mid b$ then has $w_z(b) \ge 1$, so the right derivative of the objective in z_p equals $R_p - \sum_{b \in I} \mathrm{v}_p(b) \le 0$. Hence the objective does not increase on $z_p \ge 1$. $\square$

Lemma 8.4 (rounding)**.** *There is a set $B \subseteq I$ with $\prod A \mid \prod B$ and $|B| \le \tau^* + |\mathcal{P}|$.*

Proof. Let y^* be an optimal vertex of the polytope in (4). A vertex of a polytope in $\mathbb{R}^m$ is determined by m linearly independent tight constraints, at most $|\mathcal{P}|$ of which are covering constraints, so at most $|\mathcal{P}|$ coordinates of y^* lie strictly between 0 and 1. Put $B = \{b : y_b^* > 0\}$. Then $\sum_{b \in B} \mathrm{v}_p(b) \ge \sum_b \mathrm{v}_p(b) y_b^* \ge R_p$ for every p, i.e. $\prod A \mid \prod B$, and $|B| \le \#\{b : y_b^* = 1\} + |\mathcal{P}| \le \tau^* + |\mathcal{P}|$. $\square$

Lemma 8.5 (few primes)**.** *If $m < 8n^3$ then $|\mathcal{P}| < 16n$.*

Proof. The i-th prime is at least $i+1$, so if $|\mathcal{P}| \ge 16n$ then $\prod_{p \in \mathcal{P}} p \ge (16n+1)! > (16n/e)^{16n}$, while $\prod_{p \in \mathcal{P}} p \le \prod A \le m^n < (8n^3)^n$. Taking logarithms, $16 \ln(16n/e) = 28.3\ldots + 16 \ln n > \ln 8 + 3 \ln n$ for every $n \ge 1$, a contradiction. $\square$

Proof of Theorem 8.2. If $a_n \geq 8n^3$, Theorem 1.3 gives $2n \leq 18n$ elements. Otherwise let z be a maximiser in Lemma 8.3 with $z_p \leq 1$, supported on $\mathcal{P}$. Since $w_z(a) \leq 2 + (w_z(a)-2)^+$ and $A \subseteq [1, m]$,

$$\sum_{a\in A} w_z(a) \leq 2n + \sum_{k=1}^{m} \big(w_z(k)-2\big)^+ \leq 2n + \sum_{b\in I} \big(w_z(b)-1\big)^+$$

by (3), so $\tau^* \leq 2n$. Lemmas 8.4 and 8.5 give $|B| \leq 2n + 16n - 1 < 18n$. $\square$

Equivalent forms and a per-prime-power sufficient condition

Subtracting $\sum w_z$ over both ranges, (3) is equivalent to

$$\sum_{b\in I} \min\big(w_z(b), 1\big) \leq \sum_{k=1}^{m} \min\big(w_z(k), 2\big) + \Delta, \qquad \Delta := \sum_{b\in I} w_z(b) - \sum_{k=1}^{m} w_z(k) = \sum_q z_q \delta_q \geq 0, \tag{5}$$

where q runs over prime powers, $z_{p^j} := z_p$, and $\delta_q := \#\{b \in I : q \mid b\} - \lfloor m/q \rfloor \in \{0, 1\}$. Writing $\min(w, 1) = w \min(1, 1/w)$ and $\min(w, 2) = w \min(1, 2/w)$ and expanding $w_z(n) = \sum_{q|n} z_q$, both sides of (5) become sums over prime powers, and (3) follows if for every prime power q with $z_q > 0$

$$\sum_{b\in I,\ q|b} \min\Big(1, \frac{1}{w_z(b)}\Big) \leq \delta_q + \sum_{k\leq m,\ q|k} \min\Big(1, \frac{2}{w_z(k)}\Big). \tag{6}$$

Substituting $b = qi'$, $k = qi$ and $w_z(qi) = a + w_z(i)$ with $a := w_z(q)$, the multiples of q in I correspond to an interval J of $L := \lfloor m/q \rfloor$ or $L + 1$ consecutive integers, and (6) follows from

$$\sum_{i\in J} f_a\big(w_z(i)\big) \leq \sum_{i=1}^{L} g_a\big(w_z(i)\big), \qquad f_a(u) := \min\Big(1, \frac{1}{a+u}\Big), \quad g_a(u) := \min\Big(1, \frac{2}{a+u}\Big), \tag{7}$$

for intervals J of length L (the possible extra element of J is paid by δ_q, as $f_a \leq 1$).

Proposition 8.6. *Inequality* (7) *holds for every interval J of length L and every $a \geq 0$ in each of the following cases: (i) z is supported on a single prime; (ii) $a \geq 2$ and $aL \geq w_z(L!)$, in particular whenever $a \geq \max\{2, \sum_{p\leq L} z_p/(p-1)\}$.*

Proof. (i) Let z be supported on p and $\phi(j) := f_a(z_p j)$, a nonincreasing function of j, with increments $D_j := \phi(j-1) - \phi(j) \geq 0$. For any set S of L integers, $\sum_{i\in S} \phi(\mathrm{v}_p(i)) = L\phi(0) - \sum_{j\geq 1} D_j \#\{i \in S : p^j \mid i\}$. Since an interval of length L contains at least $\lfloor L/p^j \rfloor = \#\{i \leq L : p^j \mid i\}$ multiples of p^j, the sum over J is at most the sum over $[1, L]$, and $f_a \leq g_a$ pointwise. (ii) For $a \geq 2$ the minima are inactive; the left side is at most L/a, and by the Cauchy–Schwarz inequality the right side is $2\sum_{i\leq L}(a + w_z(i))^{-1} \geq 2L^2/(aL + w_z(L!)) \geq L/a$. Finally $w_z(L!) = \sum_p z_p \mathrm{v}_p(L!) \leq L \sum_{p\leq L} z_p/(p-1)$ by Legendre's formula. $\square$

Three strengthenings that fail

Each of the following statements would imply (3) or (7) by a short argument, and each is false. The common mechanism is that an interval of length L can contain more integers free of all primes $\leq L$ than $\pi(L) + 1$: the pattern of such integers is an admissible set, and by the Chinese remainder theorem every admissible pattern occurs, so their maximal number is the

Hensley–Richards function $\rho^*(L)$, which exceeds $\pi(L)$ for $L \geq 3159$ and by an unbounded amount as $L \to \infty$ [5, 6]. Explicit windows were built by a centred sieve (residue class 0 modulo every $p \leq \sqrt{L}$, then the least populated class modulo each larger prime) and the Chinese remainder theorem; the starting points x and verification scripts are in the repository.

Proposition 8.7. *(a)* (weight-free transport) *There need not exist an injection $\varphi : [1, m] \to I$ such that for every k some prime $p \mid k$ satisfies $(k/p) \mid \varphi(k)$. For $m = 20000$ there is an x (with 4298 digits) for which I contains 2270 integers free of all primes $\leq m/2$; every k related to such an integer is 1 or a prime, so Hall's condition fails because $2270 > \pi(m) + 1 = 2263$. (Such an injection would give (3) at once, since $w_z(\varphi(k)) \geq w_z(k) - z_p \geq w_z(k) - 1$.)*

(b) (single-threshold level sets) *The inequality $\#\{i \leq L : w_z(i) \geq 2\tau\} \leq \#\{i \in J : w_z(i) \geq \tau\}$ fails for $L = 10^5$, $z_p = 1$ for all $p \leq L/2$, at $\tau = 1$ (by 142) and at $\tau = 1/2$ (by 5275), although (7) holds on the same window with room to spare. (Integrating it over τ would give (7).)*

(c) (Laplace form) *The inequality $s \sum_{i \in J} s^{w_z(i)} \leq \sum_{i \leq L} s^{w_z(i)}$ for all $s \in (0, 1]$, which would give (7) for all $a \geq 2$ by integrating against $s^{a-1}\,ds$, fails for $L = 10^6$, $z_p = 1$ for all $p \leq L$, and $s \in [0.005, 0.015]$: the window (with x of 433636 digits) contains 80436 integers free of all primes $\leq L$, against $\pi(L) + 1 = 78499$, and at $s = 0.005$ the two sides are 405.03 and 398.77.*

Remark 8.8 (evidence and the remaining programme)**.** Conjecture 8.1 was tested on 2.1×10^6 random and adversarial instances with $m \leq 600$ (0/1, fractional and $p^{-\theta}$ weights; $x = 0$, random x, $x \equiv -1$ modulo products of weighted primes, windows centred at highly divisible numbers) and, with nine weight choices, on the dense windows of Proposition 8.7 of lengths $2 \cdot 10^4$ and 10^5, where the slack is at least $0.18m$; (7) was tested on 3.5×10^5 instances over all prime powers. Equality in (3) occurs only for $x = 0$ and single primes. For 0/1 weights $z = \mathbf{1}_{\mathcal{Q}}$ the inequality is equivalent to $B_0 - A_0 \leq \Delta + B_{\geq 2}$, where B_0 and A_0 count the integers in $[1, m]$ and in I free of the primes of $\mathcal{Q}$, and $B_{\geq 2}$ counts the $k \leq m$ with at least two prime factors from $\mathcal{Q}$ (with multiplicity); by inclusion–exclusion $B_0 - A_0$ is an alternating sum of the surpluses δ_d over squarefree products d of primes of $\mathcal{Q}$. In the layer-cake form of (7) the integrand is positive only where the count on $[1, L]$ is stuck at a low level, and its positive part is bounded by the number of integers in a window free of the weighted primes, i.e. by an explicit large sieve bound, while the negative part is a counting quantity on $[1, L]$. Making this quantitative for all weights is the remaining step; the failures in Proposition 8.7 show that it cannot be replaced by a weight-free or single-threshold argument.

9 Counting certificates and the 0/1 case

All three failures in Proposition 8.7 have a common feature: they use only the lower bound $N_I(d) \geq \lfloor m/d \rfloor$ on the number $N_I(d) := \#\{b \in I : d \mid b\}$ of multiples of d in the interval. The upper bound $N_I(d) \leq \lfloor m/d \rfloor + 1$, which is exactly what the dense windows violate in spirit, was never used. Using both bounds, linear programming duality turns the hinge inequality into a finite question, and this question has a positive answer in every case we could compute; for 0/1 weights it also has an explicit answer with a slightly larger threshold.

The certificate principle

Proposition 9.1 (counting certificates)**.** *Let z be a weight and let (c_d) be real numbers, finitely many nonzero, indexed by positive integers d, such that*

$$\sum_{d|n} c_d \le \big(w_z(n)-1\big)^+ \qquad \text{for every positive integer } n. \tag{8}$$

Put $V(c) := \sum_{c_d>0} c_d\lfloor m/d\rfloor - \sum_{c_d<0} |c_d|\,(\lfloor m/d\rfloor+1)$. Then for every $x \ge 0$, $\sum_{b\in I}(w_z(b)-1)^+ \ge V(c)$. In particular (3) *holds for (m,z) and every x as soon as $V(c) \ge W(m,z) := \sum_{k\le m}(w_z(k)-2)^+$ for some c satisfying* (8)*.*

Proof. Summing (8) over $b \in I$ gives $\sum_{b\in I}(w_z(b)-1)^+ \ge \sum_d c_d N_I(d)$, and $\lfloor m/d\rfloor \le N_I(d) \le \lfloor m/d\rfloor + 1$ for every d because I consists of m consecutive integers. □

Corollary 9.2 (the trivial regime)**.** *For every weight z, every $m \ge 1$ and every $x \ge 0$, $\sum_{b\in I}(w_z(b)-1)^+ \ge \sum_{k\le m} w_z(k) - m$. Hence the hinge inequality with threshold $c \ge 1$ holds for (m,z) and every x whenever $\sum_{k\le m,\, w_z(k)<c}(c-w_z(k)) \le (c-1)m$; for 0/1 weights,* (3) *holds whenever $\#\{k \le m : w_z(k) = 0\} \le \#\{k \le m : w_z(k) \ge 2\}$, in particular for P the set of all primes up to m and every m.*

Proof. This is Proposition 9.1 for the affine certificate $c_1 = -1$, $c_{p^j} = z_p$ $(j \ge 1)$, which satisfies (8) with $\sum_{d|n} c_d = w_z(n) - 1$; directly, $\sum_I (w_z-1)^+ \ge \sum_I (w_z-1) = \sum_{p,j} z_p N_I(p^j) - m \ge \sum_{k\le m} w_z(k) - m$. Since $\sum_{k\le m}(w_z(k)-c)^+ = \sum_{k\le m} w_z(k) - cm + \sum_{w_z(k)<c}(c-w_z(k))$, the stated condition is exactly $\sum_K (w_z-c)^+ \le \sum_K w_z - m$. For 0/1 weights and $c = 2$ the condition reads $2\#\{w_z = 0\} + \#\{w_z = 1\} \le m$. For the full set of primes $\#\{w_z = 0\} = 1$ and $\#\{w_z \ge 2\} \ge 1$ for $m \ge 4$, while for $m \le 3$ the left side of (3) vanishes. □

So the hinge inequality is nontrivial only when integers of weight below the threshold dominate $[1,m]$: for 0/1 weights, prime sets without small primes (for P the primes in $[53, 653]$ and $m = 10^8$ one has $\#\{\omega_P = 0\} = 61{,}911{,}251$ against $\#\{\omega_P \ge 2\} = 7{,}944{,}229$, and the trivial bound is negative); for fractional weights, weights spread over primes of many sizes with $\sum_{k\le m} w_z(k)$ of order m. This is also the regime of the dense windows of Proposition 8.7.

The best certificate is the optimum of a linear programme: the dual of minimising $\sum_t n_t(w_z(t)-1)^+$ over nonnegative real "multisets" (n_t) of valuation patterns t subject to $\lfloor m/d\rfloor \le \sum_{t:d|t} n_t \le \lfloor m/d\rfloor + 1$ for all d. We solved this programme (for general weights with the exponents capped at $\lfloor \log_p m\rfloor + 1$, which is legitimate because a certificate supported on capped d satisfies (8) for all n; for 0/1 weights with c supported on pairs and triples of primes, adding the constraints (8) by separation) and found $V \ge W$ in every instance: for all weights $z \in \{1, 0.7, 0.4\}^P$ and mixed weights with P the primes up to 17 and $m \le 200$ (e.g. $V = 114$ against $W = 64$ for $m = 100$, $z \equiv 1$), and for 0/1 weights with P the set of all primes up to m for $m \in \{300, 500, 1000\}$ ($V = 223.5$, 407.4, 907.0 against the distinct-prime targets $W_\omega = 59$, 124, 321 defined below; these are the values of the programme restricted to pairs and triples, and the affine certificate of Corollary 9.2 alone gives 277, 507, 1125) as well as for various subsets of the primes at $m = 400$. The optimal certificates found (they are not unique) are small: $c_{p^j} = 1$ for prime powers with $j \ge 2$, $c_{pq} = 1$ for pairs containing 2, 3 or 5, and $c_{pqr} = -1$ for a few triples (no coefficient at $d = 1$ or at single primes). The scripts are in the repository.

An explicit family for 0/1 weights

For 0/1 weights $z = \mathbf{1}_P$ write $\omega_P(n)$ for the number of primes of P dividing n, $\Omega_P(n) = \sum_{p\in P} \mathrm{v}_p(n)$ and $E_P := \Omega_P - \omega_P = \sum_{p\in P}\sum_{j\geq 2}[p^j \mid n]$. Since $(\Omega_P - 1)^+ = E_P + (\omega_P - 1)^+$, $(\Omega_P - c)^+ \leq E_P + (\omega_P - c)^+$ for every $c \geq 0$, and $\sum_{b\in I} E_P(b) \geq \sum_{k\leq m} E_P(k)$ (count the multiples of each p^j), it suffices to certify the distinct-prime inequality $\sum_{k\leq m}(\omega_P(k) - 2)^+ \leq \sum_{b\in I}(\omega_P(b) - 1)^+$, whose left side we denote $W_\omega(m)$ (for $z = \mathbf{1}_P$ one has $W(m, z) \geq W_\omega(m)$, and the certificates below are for the distinct-prime version).

Lemma 9.3. *Let R be the set of the r smallest primes of P. Put $c_d := (-1)^{|A|}$ for $d = \prod A$ with $A \subseteq R$, $|A| \geq 2$, and $c_d := (-1)^{|A|+1}$ for $d = q\prod A$ with $q \in P \setminus R$ and $\emptyset \neq A \subseteq R$; all other $c_d = 0$. Then (8) holds with $(\omega_P(n) - 1)^+$ on the right, and it keeps holding if all coefficients belonging to any fixed $q \in P \setminus R$ are set to 0. For $r = 2$, $R = \{2, 3\}$ and the coefficients of the primes $q > m/2$ removed, the value is*

$$V_2(m) := \left\lfloor \frac{m}{6} \right\rfloor + \sum_{\substack{q\in P\\ 5\leq q\leq m/2}} \Big(\left\lfloor \frac{m}{2q} \right\rfloor + \left\lfloor \frac{m}{3q} \right\rfloor - \left\lfloor \frac{m}{6q} \right\rfloor - 1 \Big). \tag{9}$$

Proof. Let n have prime set $T = A_0 \cup Q_0$ with $A_0 \subseteq R$, $Q_0 \subseteq P \setminus R$. Then $\sum_{d|n} c_d = \sum_{A\subseteq A_0, |A|\geq 2}(-1)^{|A|} + \sum_{q\in Q_0}\sum_{\emptyset\neq A\subseteq A_0}(-1)^{|A|+1} = (|A_0| - 1)^+ + |Q_0|\,[A_0 \neq \emptyset] \leq (|T| - 1)^+$, using $\sum_{A\subseteq A_0}(-1)^{|A|} = [A_0 = \emptyset]$. Removing the terms of one q removes a nonnegative contribution. The value follows from the definition of V. □

With $\mathcal{A}(x) := \sum_{p\leq x}\lfloor x/p\rfloor = \sum_{n\leq x}\omega(n)$ and P the set of all primes up to m one has the exact identities $W_\omega(m) = \mathcal{A}(m) - 2m + 2 + \#\{\text{prime powers} \leq m\}$ and $V_2(m) = \mathcal{A}(m/2) + \mathcal{A}(m/3) - \mathcal{A}(m/6) - \lfloor m/4\rfloor - \lfloor m/6\rfloor - \lfloor m/9\rfloor + \lfloor m/12\rfloor + \lfloor m/18\rfloor - \pi(m/2) + 2$, so $V_2(m) \geq W_\omega(m)$ was checked exactly for every $m \leq 10^7$ (at $m = 10^3, \ldots, 10^7$ the ratio V_2/W_ω is 2.72, 2.00, 1.71, 1.55, 1.44). Since $\mathcal{A}(x) = x\ln\ln x + Mx + O(x/\ln x)$ [7, Thm. 430], $V_2(m) - W_\omega(m) = -\frac{1}{3}m\ln\ln m + (\frac{29}{18} - \frac{M}{3})m + o(m)$, so this single certificate proves the 0/1 case of (3) for the full set of primes for every $m \leq 10^7$ (checked exactly for each m) and fails for all sufficiently large m (the leading-order crossing is at $\ln\ln m = \frac{29}{6} - M \approx 4.57$, i.e. $m \approx 10^{42}$). For the full set of primes this is of course weaker than Corollary 9.2, which settles that case for every m; the point of the family is that it does not need small primes in P, where the affine certificate fails. The number of anchors must depend on P: for P the primes in $[53, 653]$ and $m = 10^8$ the two-anchor certificate gives $1{,}595{,}615 < W_\omega = 1{,}602{,}217$, while three anchors give $2{,}266{,}053$.

The 0/1 case with a larger threshold

Theorem 9.4. *Let P be a finite set of primes, $m \geq 3$, $x \geq 0$, and $c^*(m) := 5 + \max\big(2, \lceil 2e^2\ln(1 + \frac{\ln m}{6})\rceil\big)$. Then*

$$\sum_{n\leq m}\big(\Omega_P(n) - c^*(m)\big)^+ \leq \sum_{b\in I}\big(\Omega_P(b) - 1\big)^+, \tag{10}$$

and $c^*(m) \leq 20\ln\ln m$ for $m \geq 30$.

The proof rests on an elementary sieve lemma which, in the language of Proposition 9.1, is a certificate with positive coefficients on single primes and negative coefficients on pairs.

Lemma 9.5. *For $y \geq 2$, $\sum_{p\leq y} 1/p \leq e\ln(1 + \ln y)$.*

Proof. With $\sigma = 1 + 1/\ln y$ one has $p^{\sigma-1} \le e$ for $p \le y$, so $1/p \le e\, p^{-\sigma}$; and $\sum_p p^{-\sigma} \le \ln \zeta(\sigma) \le \ln(1 + 1/(\sigma - 1)) = \ln(1 + \ln y)$. □

Lemma 9.6 (sieve lemma)**.** *Let $L \ge 64$, let Q be a finite set of primes, let J be any set of L consecutive integers and $s(L) := \max(2, \lceil 2e^2 \ln(1 + \frac{\ln L}{6})\rceil)$. Then*

$$\#\{n \le L : \omega_Q(n) \ge s(L) + 5\} \le \#\{n \in J : \omega_Q(n) \ge 1\}.$$

Proof. Put $y = L^{1/6}$, $R = \{q \in Q : q \le y\}$, $H = \sum_{q\in R} 1/q$ and $s = s(L)$. An integer $n \le L$ has at most five prime factors exceeding y, so $\omega_Q(n) \ge s + 5$ forces $\omega_R(n) \ge s$, and $N_s := \#\{n \le L : \omega_R(n) \ge s\} \le \sum_{S\subseteq R, |S|=s} \lfloor L/\prod S\rfloor \le L\, H^s/s!$ by the bound $s!\, e_s \le H^s$ on elementary symmetric functions. By Lemma 9.5, $H \le e \ln(1 + \frac{\ln L}{6})$, so $s \ge 2eH$ and, with $s! \ge (s/e)^s$, $N_s \le L(eH/s)^s \le L2^{-s} \le L/4$.

For the right-hand side we use that any set J of L consecutive integers satisfies $L/d - 1 < N_J(d) < L/d + 1$. If $H \ge 1/2$, add primes of R one at a time until the reciprocal sum first reaches $1/2$; the resulting R' has $1/2 \le H' \le 1$ (the last prime added contributes at most $1/2$) and $t' = |R'| \le y$. By the two-term Bonferroni inequality, $\#\{n \in J : \omega_Q(n) \ge 1\} \ge \#\{n \in J : \omega_{R'}(n) \ge 1\} \ge \sum_{q\in R'} N_J(q) - \sum_{q<q'} N_J(qq') \ge L(H' - H'^2/2) - t'(t' + 1)/2 \ge \frac{3}{8}L - \frac{3}{4}y^2 \ge \frac{3}{8}L - \frac{1}{8}L = \frac{1}{4}L \ge N_s$, using $y(y + 1)/2 \le \frac{3}{4}y^2$ for $y \ge 2$ and $\frac{3}{4}L^{1/3} \le L/8$ for $L \ge 64$. If $0 < H < 1/2$, the same estimate with $R' = R$ and $H - H^2/2 \ge \frac{3}{4}H$ gives $\#\{n \in J : \omega_R(n) \ge 1\} \ge \frac{3}{4}LH - \frac{3}{4}y^2 \ge \frac{1}{4}LH$ (because $H \ge 1/y$ and $\frac{3}{4}y^2 \le L/(2y)$ for $L \ge 64$), while $N_s \le LH^2/2 \le LH/4$. If $H = 0$ then $N_s = 0$. □

Proof of Theorem 9.4. Order P increasingly and let $\omega_{<p}(n)$ count the primes of P below p dividing n. For every integer $c \ge 0$,

$$(\omega_P(n) - c)^+ = \sum_{p\in P,\ p|n} [\omega_{<p}(n) \ge c], \tag{11}$$

since the right side counts all but the c smallest primes of P dividing n. Fix $p \in P$ and put $L = \lfloor m/p\rfloor$. The multiples $n = pi \le m$ of p have $\omega_{<p}(pi) = \omega_{<p}(i)$ with $i \le L$, and the multiples $b = pi$ of p in I have i ranging over a set J_p of L or $L + 1$ consecutive integers. If $L \ge 64$, Lemma 9.6 applied to $Q = \{q \in P : q < p\}$ and to L consecutive elements of J_p gives $\#\{i \le L : \omega_{<p}(i) \ge c^*(m)\} \le \#\{i \in J_p : \omega_{<p}(i) \ge 1\}$, because $c^*(m) \ge s(L) + 5$. If $L < 64$ the left side is 0, as $\omega_{<p}(i) \le 3 < 7 \le c^*(m)$ for $i \le 63$. Summing over p and using (11) with $c = c^*(m)$ on the left and $c = 1$ on the right proves (10) with ω_P in place of Ω_P; the peeling identities stated before Lemma 9.3 then give (10). Finally $G(t) := 20 \ln t - 7 - 2e^2 \ln(1 + t/6)$ is increasing for $t > 0$ (as $2e^2 < 20$) and $G(\ln 30) > 0$, which gives $c^*(m) \le 20 \ln \ln m$ for $m \ge 30$. □

Remark 9.7 (fractional weights)**.** Theorem 9.4 does not imply (3) for fractional weights, and integrating it over weight thresholds cannot (Proposition 8.7 and the identity $\int_0^1(\#\{p \mid n : z_p > t\} - 1)^+dt = \sum_{p|n} z_p - \max_{p|n} z_p$). Writing $w_z = E + S$ with $S(n) = \sum_p \min(z_p\, \mathrm{v}_p(n), 1)$ and $E = \sum_p(z_p\, \mathrm{v}_p(n) - 1)^+$, one has $(w_z - 1)^+ = E + (S - 1)^+$ (if $S < 1$ then every $z_p\, \mathrm{v}_p < 1$, so $E = 0$), $(w_z - c)^+ \le E + (S - c)^+$, and $\sum_I E \ge \sum_K E$ because $(z_p\, \mathrm{v}_p(n) - 1)^+ = \sum_{j\ge1} \Delta_{p,j}[p^j \mid n]$ with nonnegative increments $\Delta_{p,j} = (z_p j - 1)^+ - (z_p(j - 1) - 1)^+$, so (3) for all weights follows from its analogue for the bounded function S; for rational weights $z_p = a_p/N$ this is the integer statement $\sum_{k\le m}(R(k) - 2N)^+ \le \sum_{b\in I}(R(b) - N)^+$ with $R(n) = \sum_p \min(a_p\, \mathrm{v}_p(n), N)$, of which the 0/1 case ($a_p \in \{0, N\}$) is the one treated above. The linear programme of Proposition 9.1 certifies the fractional inequality in every instance we solved; an explicit certificate for all m and all weights would complete the proof of $g(n) \le 18n$.

10 The fractional case and an $O(n \ln\ln n)$ bound

Theorem 9.4 used only 0/1 weights. In this section the same method, combined with a random-thinning device, handles arbitrary weights, at the price of a larger threshold; through Lemma 8.3 this gives the first bound for $g(n)$ that is linear up to a $\ln\ln n$ factor.

Theorem 10.1. *For every weight z, every $m \ge 16$ and every $x \ge 0$,*

$$\sum_{k \le m} \big(w_z(k) - c_m\big)^+ \le \sum_{b \in I} \big(w_z(b) - 1\big)^+, \qquad c_m := 1 + 149e \ln(1 + \ln m), \tag{12}$$

and $c_m < 675 \ln\ln m$. For $2 \le m \le 15$ the inequality holds with $c_m = 4$.

Corollary 10.2. *$g(n) \le \big(17 + 149e \ln(1 + \ln(8n^3))\big)n$ for all $n \ge 1$; in particular $g(n) = O(n \ln\ln n)$.*

Proof. If $a_n \ge 8n^3$, Theorem 1.3 gives $2n$. Otherwise $m = a_n < 8n^3$ and, by Lemma 8.3 with a maximiser z, $\sum_{a \in A} w_z(a) \le c_m n + \sum_{k \le m} (w_z(k) - c_m)^+ \le c_m n + \sum_{b \in I} (w_z(b) - 1)^+$, so $\tau^* \le c_m n$ and Lemmas 8.4, 8.5 give $|B| < (c_m + 16)n$, with $c_m = 4$ when $m \le 15$; in all cases $c_m \le c_{8n^3}$ since c_m is increasing for $m \ge 16$ and $4 < c_{16}$. □

Throughout this section $\lambda(t) := \ln(1 + \ln t)$, $L \ge 16$ is an integer, J is a set of L consecutive positive integers, $R := \lfloor \log_2 L \rfloor$, $H := e\lambda(L)$ and $L_0 := \ln(H + 2)$; note $H > 3$ and $L_0 > 1$ for $L \ge 16$.

Two-term Bonferroni at level k

Lemma 10.3. *For integers $r \ge 0$ and $k \ge 1$, $[r \ge k] \ge \binom{r}{k} - k\binom{r}{k+1}$.*

Proof. For $r < k$ both sides vanish; for $r \in \{k, k+1\}$ the right side is 1; for $r \ge k + 2$, $\binom{r}{k} - k\binom{r}{k+1} = \binom{r}{k}\big(1 - \frac{k(r-k)}{k+1}\big) \le 0$. □

A sieve lemma with an arbitrary gain

Let D be a finite family of pairwise coprime integers $d \ge 2$ and $\nu_D(n) := \#\{d \in D : d \mid n\}$. For $1 \le k \le R - 1$ put

$$\rho(k) := \left\lceil 16H + \frac{16kL_0}{\sqrt{\ln(2 + k/H)}} \right\rceil, \qquad C(k) := 2k + \rho(k). \tag{13}$$

Lemma 10.4 (gain-G sieve)**.** *For every real $1 \le G \le 2H$, every $1 \le k \le R - 1$ and every J,*

$$G \cdot \#\{n \le L : \nu_D(n) \ge C(k)\} \le \#\{n \in J : \nu_D(n) \ge k\}.$$

Proof. Put $y := L^{1/(k+1)} \geq 2$, $D_0 := \{d \in D : d \leq y\}$ and $H_0 := \sum_{d\in D_0} 1/d$. Choosing a prime divisor of each $d \in D_0$ gives distinct primes $\leq y$, so $H_0 \leq \sum_{p\leq y} 1/p \leq e\lambda(y) \leq H$ by Lemma 9.5. Let e_j denote the elementary symmetric functions of the numbers $1/d$, $d \in D_0$; then $(k+1)e_{k+1} \leq H_0 e_k$ and $e_{k+r} \leq e_k e_r / \binom{k+r}{k} \leq e_k H_0^r / r!$.

Lower bound on J. Retain each element of D_0 independently with probability $\theta := 1/(8H)$, obtaining a random subfamily E. For every E, Lemma 10.3 summed over $n \in J$ gives $\#\{n \in J : \nu_E(n) \geq k\} \geq M_k(E) - kM_{k+1}(E)$, where $M_j(E) := \sum_{F\subseteq E, |F|=j} N_J(\prod F)$; since $\nu_D \geq \nu_E$ and $\mathbb{E}\, M_j(E) = \theta^j M_j(D_0)$, taking expectations yields $\#\{n \in J : \nu_D(n) \geq k\} \geq \theta^k \big(M_k(D_0) - k\theta M_{k+1}(D_0)\big)$. A product of k elements of D_0 is at most $y^k = L/y \leq L/2$, so $N_J(a) \geq L/a - 1 \geq L/(2a)$ and $M_k(D_0) \geq \frac{L}{2} e_k$; a product of $k+1$ elements is at most $y^{k+1} = L$, so $N_J(a) \leq L/a + 1 \leq 2L/a$ and $M_{k+1}(D_0) \leq 2Le_{k+1}$. Hence $k\theta M_{k+1}(D_0) \leq 2\theta L H_0 e_k \leq \frac{L}{4} e_k$ and

$$\#\{n \in J : \nu_D(n) \geq k\} \geq \frac{Le_k}{4(8H)^k}. \tag{14}$$

Upper bound on $[1, L]$. An integer $n \leq L$ is divisible by at most k elements of $D \setminus D_0$, since $k+1$ of them would have product $> y^{k+1} = L$. Hence $\nu_D(n) \geq C(k)$ implies $\nu_{D_0}(n) \geq k + \rho(k)$, and with $r = \rho(k)$,

$$\#\{n \leq L : \nu_D(n) \geq C(k)\} \leq Le_{k+r} \leq Le_k \frac{H^r}{r!}. \tag{15}$$

The gain. Write $s = k/H$ and $\ell = \ln(2+s)$. We claim $\ln\big(\frac{16}{e}(1+s/\sqrt{\ell})\big) \geq \frac{1}{2}\sqrt{\ell}$. With $t = \sqrt{\ell}$: if $t \leq 1$ the left side is at least $\ln(16/e) > 1 \geq t/2$; if $t \geq 1$ then $s = e^{t^2} - 2$ and $1 + s/t \geq e^{t^2}/(2t)$ (as $e^{t^2}/2 + t - 2 \geq 0$), so the difference of the two sides is at least $\ln(8/e) + t^2 - \ln t - t/2$, which is positive at $t = 1$ and increasing. Since $r = \rho(k) \geq 16H(1 + s\, L_0/\sqrt{\ell}) \geq 16H(1 + s/\sqrt{\ell})$, we get $\ln \frac{r}{eH} \geq \frac{1}{2}\sqrt{\ell}$ and, using $r! \geq (r/e)^r$ and $r \geq 16kL_0/\sqrt{\ell}$, $\ln \frac{r!}{H^r} \geq r \ln \frac{r}{eH} \geq 8kL_0$. On the other hand $4G \leq 8H \leq (H+2)^3$ gives $\ln\big(4G(8H)^k\big) \leq 3(k+1)L_0 \leq 6kL_0$. Therefore $4G(8H)^k H^r/r! \leq 1$, and (14), (15) give the lemma. $\square$

Dyadic encoding of fractional valuations

Let $u = (u_q)$ be weights, $s_q(n) := \min(u_q \,\mathrm{v}_q(n), 1)$ and $X(n) := \sum_q s_q(n)$. For $j \geq 0$ put $N_j(n) := \#\{q : s_q(n) \geq 2^{-j}\}$ and, for $u_q > 0$, $d_{q,j} := q^{\lceil 2^{-j}/u_q \rceil}$. Then $s_q(n) \geq 2^{-j}$ if and only if $d_{q,j} \mid n$, the integers $d_{q,j}$ (q varying) are pairwise coprime, so $N_j = \nu_{D_j}$ for $D_j := \{d_{q,j}\}$; and since $s \leq \sum_{j\geq 0} 2^{-j}[s \geq 2^{-j}] \leq 2s$ for $s \in [0, 1]$,

$$X(n) \leq \sum_{j\geq 0} 2^{-j} N_j(n) \leq 2X(n). \tag{16}$$

Lemma 10.5 (weighted sieve lemma)**.** *For every family of weights u, every $L \geq 16$ and every J,*

$$\#\{n \leq L : X(n) \geq 149H\} \leq \#\{n \in J : X(n) \geq 1\}.$$

Proof. Put $Q := 1 + \lfloor \log_2(R-1) \rfloor$, so that $2^j \leq R - 1$ for $j < Q$ and $2^Q \geq R$. With $t = \ln L$ one has $R < 2t$ and $Q < 3 + 2\ln t < 5\lambda(L) < 2H$ for $L \geq 16$. For $j < Q$ set $a_j := C(2^j) - 1$ and for $j \geq Q$ set $a_j := R$; let $A := \sum_{j\geq 0} 2^{-j} a_j$. Since $N_j(n) \leq \omega(n) \leq R$ for $n \leq L$, if $X(n) \geq A + 1$ for some $n \leq L$ then by (16) some $j < Q$ has $N_j(n) \geq C(2^j)$. Lemma 10.4 with $D = D_j$, $k = 2^j$ and $G = Q$ gives $Q \cdot \#\{n \leq L : N_j(n) \geq C(2^j)\} \leq \#\{n \in J : N_j(n) \geq 2^j\} \leq \#\{n \in J : X(n) \geq 1\}$, the last step because $N_j(n) \ge 2^j$ forces $X(n) \ge 2^{-j}N_j(n) \ge 1$. Summing over the Q indices $j < Q$, $\#\{n \le L : X(n) \ge A+1\} \le \#\{n \in J : X(n) \ge 1\}$.

It remains to show $A+1 \le 149H$. From (13), $a_j \le 2\cdot 2^j + 16H + 16\cdot 2^j L_0/\sqrt{\ln(2+2^j/H)}$, so $\sum_{j<Q} 2^{-j}a_j \le 2Q + 32H + 16L_0 S_Q$ with $S_Q := \sum_{j<Q}\ln(2+2^j/H)^{-1/2}$. The indices with $2^j \le H$ number at most $1+\log_2 H \le 1+\frac{3}{2}\sqrt{H}$ and contribute at most $\frac{5}{4}$ each, hence at most $\frac{21}{8}\sqrt{H}$; if j_0 is the first index with $2^{j_0} > H$, the j_0-term is less than 1 and the term j_0+t $(t \ge 1)$ is less than $\frac{5}{4}t^{-1/2}$, so the remaining terms contribute less than $1+\frac{5}{2}\sqrt{Q} \le \frac{21}{5}\sqrt{H}$. Thus $S_Q \le 7\sqrt{H}$, and with $L_0 = \ln(H+2) \le \sqrt{H}$ we get $16L_0S_Q \le 112H$. Finally $\sum_{j \ge Q} 2^{-j}R = 2R/2^Q \le 2$, so $A \le 4H + 32H + 112H + 2 = 148H + 2$ and $A+1 \le 149H$ because $H > 3$. □

Proof of Theorem 10.1

Write $w_z = E + S$ with $S(n) = \sum_p \min(z_p\,\mathrm{v}_p(n), 1)$ as in the remark after Theorem 9.4; it suffices to prove $\sum_{k\le m}(S(k)-c_m)^+ \le \sum_{b\in I}(S(b)-1)^+$. By continuity in the (finitely many) weights we may assume $z_p = a_p/N$ rational; put $r_p(n) := \min(a_p\,\mathrm{v}_p(n), N)$ and $R(n) := \sum_p r_p(n) = NS(n)$. For integers n and thresholds $T \ge 0$ let $R_{<p}(n) := \sum_{q<p} r_q(n)$. Ordering the primes dividing n and letting p^* be the first with $R_{\le p^*}(n) \ge T$, one has $\sum_{p|n} r_p(n)[R_{<p}(n) \ge T] = R(n) - R_{\le p^*}(n)$ and $T \le R_{\le p^*}(n) < T + N$, whence

$$\big(R(n) - T - N\big)^+ \le \sum_{p|n} r_p(n)\big[R_{<p}(n) \ge T\big] \le \big(R(n) - T\big)^+. \tag{17}$$

Applying (17) with $T = (c_m - 1)N$ on $[1, m]$ and with $T = N$ on I, it suffices to show, for every prime p, $\sum_{k\le m,\,p|k} r_p(k)[R_{<p}(k) \ge (c_m-1)N] \le \sum_{b\in I,\,p|b} r_p(b)[R_{<p}(b) \ge N]$. Writing $r_p(n) = \sum_{j\ge 1}\Delta_{p,j}[p^j \mid n]$ with $\Delta_{p,j} = \min(a_pj, N) - \min(a_p(j-1), N) \ge 0$ and using $R_{<p}(p^ji) = R_{<p}(i)$, this follows from: for every $j \ge 1$, $L := \lfloor m/p^j \rfloor$ and every run J of L consecutive integers (the quotients i of the multiples $p^ji \in I$ form a run of L or $L+1$ consecutive integers),

$$\#\{i \le L : R_{<p}(i) \ge (c_m-1)N\} \le \#\{i \in J : R_{<p}(i) \ge N\}.$$

Dividing by N, $R_{<p}/N$ is the function X of Lemma 10.5 for the weights $u_q = z_q$ $(q < p)$. If $L \ge 16$ the lemma applies because $c_m - 1 = 149e\lambda(m) \ge 149e\lambda(L)$; if $L < 16$ the left side is empty since $X(i) \le \omega(i) \le 2 < c_m - 1$ for $i \le 15$. This proves (12) for $m \ge 16$; for $2 \le m \le 15$, $S(k) \le \omega(k) \le 2$ gives the statement with $c_m = 4$. Finally, for $m \ge 16$ and $t = \ln m$, $\lambda(m) = \ln\ln m + \ln(1+1/t) < \frac{3}{2}\ln\ln m$ (as $1/t < 3/8 < \frac{1}{2}\ln\ln m$), and $1 + 149e\lambda(m) < 150e\lambda(m) < 675\ln\ln m$. □

Remark 10.6. The constant $149e \approx 405$ is far from optimal: the thinning probability $1/(8H)$, the gain $G = Q$ and the dyadic budget were chosen for simplicity, and the bound of Corollary 10.2 is smaller than that of Theorem 1.2 only for astronomically large n. Section 11 optimises these choices and reaches the threshold $20\ln\ln m$; the resulting bound $(16 + 20\ln\ln(8n^3))n$ is below Theorem 1.2 for every n and below Theorem 1.1 for $n \gtrsim 10^{35}$. The hinge inequality with threshold 2 (Conjecture 8.1), which would give $18n$, remains open.

11 An improved constant: threshold $20\ln\ln m$

The dyadic budget of Section 10 pays about eH at every level. Replacing the dyadic levels by a geometric sequence with ratio close to 1, and allowing an arbitrary gain in Lemma 10.4, the levels can be made to share one payment of eH, and the constant $149e \approx 405$ drops to 20.

Theorem 11.1. *For every weight z, every $m \geq 3$ and every $x \geq 0$,*

$$\sum_{k \leq m} \big(w_z(k) - 20 \ln\ln m\big)^+ \leq \sum_{b \in I} \big(w_z(b) - 1\big)^+, \tag{18}$$

and for $m \leq 2$ the inequality holds with threshold 2. Consequently $g(n) \leq \big(16 + 20 \ln\ln(8n^3)\big)n$ for all $n \geq 1$.

The consequence follows exactly as in Corollary 10.2 ($20 \ln\ln m$ is increasing and exceeds 2 for $m \geq 3$). We keep the notation of Section 10: $L \geq 16$, J a set of L consecutive positive integers, $R = \lfloor \log_2 L \rfloor$, and now $h := \ln(1 + \ln L)$, $H := eh$, $h_0 := eH$. For a family D of pairwise coprime integers ≥ 2 and $1 \leq k \leq R - 1$ we use $y = L^{1/(k+1)}$, $D_0 = \{d \in D : d \leq y\}$ and the elementary symmetric functions e_r of $\{1/d : d \in D_0\}$ as before; Lemma 9.5 gives $\sum_{d \in D_0} 1/d \leq H$.

Lemma 11.2 (arbitrary gain)**.** *Let $G \geq 1$, $A := \ln(4G) + k \ln(8H)$, $s(A) := \min\{A,\ 2A/\ln(2 + A/h_0)\}$ and $\rho := \lceil h_0 + s(A) \rceil$. Then*

$$G \cdot \#\{n \leq L : \nu_D(n) \geq 2k + \rho\} \leq \#\{n \in J : \nu_D(n) \geq k\}.$$

Proof. By (14) and (15) it suffices that $\rho! \geq 4G(8H)^k H^\rho$, and by $\rho! \geq (\rho/e)^\rho$ it suffices that $\rho \ln \frac{\rho}{eH} \geq A$. Put $u = A/h_0$. For the first candidate $\rho = h_0(1+u)$ this reads $(1+u)\ln(1+u) \geq u$, which holds. For the second, $\rho = h_0(1+v)$ with $v = 2u/\ln(2+u)$, we need $(1+v)\ln(1+v) \geq u$. If $\ln(2+u) \leq 2$ then $v \geq u$ and we are done. Otherwise put $y = 2+u \geq e^2$, so $v = 2(y-2)/\ln y$; the function $2(y-2)/(\sqrt{y}\ln y)$ is increasing for $y \geq e^2$ (its logarithmic derivative is $\frac{1}{y-2} - \frac{1}{2y} - \frac{1}{y \ln y} > 0$) and equals $1.98\ldots$ at $y = e^2$, hence $v \geq \sqrt{y}$, $\ln(1+v) \geq \frac{1}{2}\ln y$ and $(1+v)\ln(1+v) \geq v \cdot \frac{1}{2}\ln y = y - 2 = u$. Rounding up preserves the inequality, since $\rho \ln(\rho/h_0)$ is increasing for $\rho \geq h_0$. □

Geometric encoding

Fix $q := 51/50$, $\delta := 1/50$, and put $k_j := \lceil q^j \rceil$; let Q be the number of $j \geq 0$ with $k_j \leq R - 1$ and $\alpha_j := \delta q^{-j}$. For weights $u = (u_p)$, $X(n) = \sum_p \min(u_p \,\mathrm{v}_p(n), 1)$ and $D_j := \{p^{\lceil q^{-j}/u_p \rceil} : u_p > 0\}$ (pairwise coprime), one has $\min(u_p \,\mathrm{v}_p(n), 1) \geq q^{-j}$ iff $p^{\lceil q^{-j}/u_p \rceil} \mid n$, so $N_j := \nu_{D_j}$ counts the primes with $\min(u_p \,\mathrm{v}_p(n), 1) \geq q^{-j}$. For $a \in (0, 1]$ and j_0 the least j with $q^{-j} \leq a$, $\sum_{j \geq j_0} \delta q^{-j} = q^{1-j_0} \geq a$; since $n \leq L$ has at most R distinct prime factors, the tail $j \geq Q$ contributes at most Rq^{1-Q}, and therefore

$$X(n) \leq \sum_{j<Q} \alpha_j N_j(n) + Rq^{1-Q} \qquad (n \leq L), \qquad \text{while} \qquad N_j(n) \geq k_j \;\Rightarrow\; X(n) \geq k_j q^{-j} \geq 1. \tag{19}$$

Lemma 11.3. *Let $h \geq 4$, $\rho_j := \rho(H, Q, k_j)$ as in Lemma 11.2 with $G = Q$, and $B_L := \sum_{j<Q} \alpha_j(2k_j + \rho_j)$. Then $\#\{n \leq L : X(n) \geq B_L + 2\} \leq \#\{n \in J : X(n) \geq 1\}$.*

Proof. Since $k_Q = \lceil q^Q \rceil \geq R$ we have $q^Q > R - 1$, and $h \geq 4$ gives $R \geq 77$, so $Rq^{1-Q} < qR/(R-1) < 1.05$. If $n \leq L$ has $N_j(n) \leq 2k_j + \rho_j - 1$ for all $j < Q$, then by (19) $X(n) < B_L - \sum_{j<Q} \alpha_j + 2 < B_L + 2$. Hence $X(n) \geq B_L + 2$ forces $N_j(n) \geq 2k_j + \rho_j$ for some $j < Q$; Lemma 11.2 with $D = D_j$, $k = k_j$, $G = Q$ and (19) give $Q \cdot \#\{n \leq L : N_j(n) \geq 2k_j + \rho_j\} \leq \#\{n \in J : N_j(n) \geq k_j\} \leq \#\{n \in J : X(n) \geq 1\}$, and summing over the Q levels proves the claim. □

Lemma 11.4 (budget)**.** *For $h \geq 4$, $B_L + 3 < 19.24\,h$.*

Proof. Write $\ell := \ln(8H) = \ln(8eh)$ and $A_j := \ln(4Q) + k_j\ell$, so $\rho_j \leq h_0 + s(A_j) + 1$. Since $R < eh/\ln 2$ and $Q \leq 1 + \ln R/\ln q$, one has $Q < 56h$; also $\sum_{j<Q} \alpha_j < q$ and, as $k_j \leq q^j + 1$, $2\sum_{j<Q} \alpha_j k_j \leq 2\delta Q + 2q < 2.24h + 2.04$, while $\sum_{j<Q} \alpha_j(h_0+1) < q(e^2h+1) < 7.538h + 1.02$. The function $f(t) = \min\{t, 2t/\ln(2+t/h_0)\}$ has $f(t)/t$ nonincreasing, hence is subadditive, so $s(A_j) \leq \ln(4Q) + k_j\ell\, g(k_j)$ with $g(u) := \min\{1, 2/\ln(2+\beta u)\}$, $\beta := \ell/(e^2h)$. The first part contributes at most $q\ln(4Q) < 1.02\ln(224h) < 1.734h$. For the second, g is decreasing and $k_j \leq q^j+1$, so $\sum_{j<Q} \alpha_j k_j g(k_j) \leq q+\delta+\frac{\delta}{\ln q}\int_1^R g(u)\,\frac{du}{u}$ with $\delta/\ln q < 1.01$; splitting the integral at $u = v_0/\beta$, $v_0 := e^2-2$, and using $\ln(2+v) \geq \ln v$ gives $\int_1^R g(u)\,du/u \leq J(h) := \ln\frac{v_0 e^2 h}{\ell} + 2\ln\frac{h-3/2}{\ln v_0}$ (because $\ln(\beta R) < h - 3/2$ for $h \geq 4$). The one-variable inequality $\ell J(h)/h < 5$ holds for $h \geq 8$ analytically ($J(h) < 3\ln h + 1.01$, $\ell < \ln h + 3.08$, and $(y+3.08)(3y+1.01)e^{-y}$ is decreasing in $y = \ln h$ with value < 4.68 at $y = \ln 8$) and on $[4,8]$ by an interval computation with 80 bins, using that ℓ and J are increasing (maximum upper bound 4.966, attained on $[4.40, 4.45]$; the true maximum is 4.910 at $h = 4.513$; script in the repository). Hence $\ell\sum_{j<Q}\alpha_j k_j g(k_j) < 6.212h$. Adding the pieces, $B_L < (7.538 + 2.240 + 1.734 + 6.212)h + 3.06$, and $3.06 + 3 < 1.515h$ for $h \geq 4$, so $B_L + 3 < 19.239h$. □

Proof of Theorem 11.1. Lemmas 11.3 and 11.4 give, for every L with $\ln(1+\ln L) \geq 4$ and every J,

$$\#\{n \leq L : X(n) \geq 19.25\,h_L - 1\} \leq \#\{n \in J : X(n) \geq 1\}, \qquad h_L := \ln(1+\ln L). \tag{20}$$

Now let m satisfy $h_m := \ln(1+\ln m) \geq 4$, put $T := 20\ln\ln m - 1$ and argue as in the proof of Theorem 10.1: with $S = \sum_p \min(z_p\,\mathrm{v}_p, 1)$, the weighted ordering identity (17) (in the normalised form with $r_p \in [0,1]$ and $N = 1$) reduces (18) for S to the transfer inequality $\#\{i \leq L_d : R_{<p}(i) \geq T\} \leq \#\{i \in J_d : R_{<p}(i) \geq 1\}$ for every prime power $d = p^j$, $L_d = \lfloor m/d \rfloor$ and every run J_d of L_d consecutive integers. Since $\ln\ln m = h_m + \ln(1 - e^{-h_m}) \geq 0.995\,h_m$ for $h_m \geq 4$, we have $T \geq 19.9h_m - 1 \geq 19.25h_{L_d} - 1$ whenever $h_{L_d} \geq 4$, and (20) applies; if $h_{L_d} < 4$ then $R_{<p}(i) \leq \omega(i) \leq \log_2 L_d < (e^4-1)/\ln 2 < 77.4 < T$, so the left side is empty. Peeling $E = w_z - S$ as before gives (18) for $h_m \geq 4$. If $m \geq 4$ and $h_m < 4$, then $\ln m < e^4 - 1$ and $w_z(k) \leq \Omega(k) \leq \log_2 m \leq 20\ln\ln m$ (the function $t/\ln t$ on $[\ln 4, e^4-1]$ is at most $13.47 < 20\ln 2$), so the left side of (18) vanishes; for $m = 3$, $20\ln\ln 3 > 1 \geq w_z(k)$; for $m \leq 2$ the left side vanishes with threshold 2. □

Remark 11.5. The constant is not sharp: with the exact ρ_j one finds $\sup_{h\geq 4}(B_L+3)/h = 17.90$, so the same argument supports a threshold of about $18\ln\ln m$; the binding steps are the bounds $\ln(224h) < 1.7h$ and $1.04\,\ell(4)/4 < 1.162$ at $h = 4$. The remaining $\ln\ln m$ comes from the baseline $\rho \geq eH$ in Lemma 11.2, i.e. from comparing the tail of ν_D on $[1, L]$ above its typical size with the count at level k in J. An absolute constant would require a comparison at the typical level; the counting certificates of Section 9 suggest that this is possible, but we have not proved it.

12 The 0/1 case with an absolute threshold

The threshold $20\ln\ln m$ of Theorem 9.4 can be replaced by the constant 4, for every prime set, by two explicit counting certificates. Throughout this section P is a finite set of primes,

$\Omega_P(n) = \sum_{p\in P} \mathrm{v}_p(n)$, $\omega_P(n) = \#\{p \in P : p \mid n\}$, $e_P := \Omega_P - \omega_P = \sum_{p\in P}(\mathrm{v}_p - 1)^+$, and

$$y := \lfloor m^{1/3} \rfloor, \qquad S := \{p \in P : p \le y\}, \qquad L := P \setminus S, \qquad H := \sum_{p\in S} \frac{1}{p}, \qquad \eta := \Big(1 + \frac{1}{y}\Big)H.$$

Theorem 12.1. *For every finite set of primes P, every $m \ge 1$ and every $x \ge 0$,*

$$\sum_{k\le m} \big(\Omega_P(k) - 4\big)^+ \le \sum_{b\in I} \big(\Omega_P(b) - 1\big)^+. \tag{21}$$

The proof uses Proposition 9.1 with the weight $z = \mathbf{1}_P$ (so $w_z = \Omega_P$), and splits according to whether $\eta \le 2$ or $\eta > 2$. We write $A := \sum_{p\in P} \sum_{j\ge 2,\, p^j \le m} \lfloor m/p^j \rfloor = \sum_{k\le m} e_P(k)$ and, for $j = 2, 3$, $E_j := \sum_{B\subseteq S,\, |B|=j} \lfloor m/\prod B\rfloor = \sum_{k\le m} \binom{\omega_S(k)}{j}$.

Lemma 12.2 (cube-root peeling)**.** *Every $k \le m$ has $\omega_L(k) \le 2$, and $(\Omega_P(k) - 4)^+ \le e_P(k) + (\omega_S(k) - 2)^+$. Hence $\sum_{k\le m}(\Omega_P(k) - 4)^+ \le A + \sum_{k\le m}(\omega_S(k) - 2)^+$.*

Proof. Three distinct primes larger than y have product at least $(y+1)^3 > m$. Thus $\Omega_P(k) - 4 = e_P(k) + \omega_S(k) + \omega_L(k) - 4 \le e_P(k) + \omega_S(k) - 2$, and $(a + b)^+ \le a + b^+$ for $a \ge 0$. ☐

Lemma 12.3 (sparse branch)**.** *Suppose $\eta \le 2$. The certificate $c_{p^j} = 1$ ($p \in P$, $j \ge 2$, $p^j \le m$), $c_{pq} = \frac{11}{21}$ ($p < q$ in S), $c_{pqr} = -\frac{1}{7}$ ($p < q < r$ in S), all other $c_d = 0$, satisfies* (8) *for $w = \Omega_P$ and has $V(c) \ge \sum_{k\le m}(\Omega_P(k) - 4)^+$.*

Proof. Feasibility. Let $r = \omega_S(n)$ and $t = \omega_P(n) \ge r$. The pair and triple coefficients contribute $\phi(r) = \frac{11}{21}\binom{r}{2} - \frac{1}{7}\binom{r}{3} = r(r-1)(13-r)/42$, and $r(13-r) \le 42$ for every integer r (with equality at $r = 6, 7$), so $\phi(r) \le (r-1)^+ \le (t-1)^+$. The prime-power coefficients contribute at most $e_P(n)$. If $t \ge 1$ then $e_P(n) + (t-1) = \Omega_P(n) - 1$; if $t = 0$ the sum is 0.

Value. Here $V(c) = A + \frac{11}{21}E_2 - \frac{1}{7}\big(E_3 + \binom{|S|}{3}\big)$. Every triple in S has product at most $y^3 \le m$, so $\binom{|S|}{3} \le E_3$. Next, $3E_3 = \sum_{\{p,q\}\subseteq S} \sum_{r\in S\setminus\{p,q\}} \lfloor m/pqr\rfloor \le H \sum_{\{p,q\}\subseteq S} m/pq$, and for $p, q \in S$ one has $pq \le y^2$, hence $\lfloor m/pq\rfloor \ge y$ and $m/pq < \lfloor m/pq\rfloor + 1 \le (1 + 1/y)\lfloor m/pq\rfloor$; therefore $3E_3 \le \eta E_2 \le 2E_2$. Consequently

$$V(c) \ge A + \tfrac{11}{21}E_2 - \tfrac{2}{7}E_3 \ge A + \Big(\tfrac{11}{21} - \tfrac{2}{7}\cdot\tfrac{2}{3}\Big)E_2 = A + \tfrac{1}{3}E_2.$$

Finally $(r-2)^+ \le \frac{1}{3}\binom{r}{2}$ for all integers $r \ge 0$ (the difference is $(r-3)(r-4)/6$ for $r \ge 3$), so $\sum_{k\le m}(\omega_S(k) - 2)^+ \le \frac{1}{3}E_2$, and Lemma 12.2 finishes the proof. ☐

Lemma 12.4 (dense branch)**.** *Suppose $\eta > 2$. The affine certificate $c_1 = -1$, $c_{p^j} = 1$ ($p \in P$, $j \ge 1$, $p^j \le m$) satisfies* (8) *for $w = \Omega_P$ and has $V(c) \ge \sum_{k\le m}(\Omega_P(k) - 4)^+$.*

Proof. Feasibility is Corollary 9.2. The value is $V(c) = \sum_{k\le m} \Omega_P(k) - (m+1)$, and since $\Omega - (\Omega - 4)^+ = \min(\Omega, 4)$ it suffices to show $\sum_{k\le m} \min(\Omega_P(k), 4) \ge m + 1$. For $y \le 4$ one has $\eta \le 0, \frac{3}{4}, \frac{10}{9}, \frac{25}{24}$ respectively, so $y \ge 5$ and $H > 2y/(y+1) \ge 5/3$. Adding the primes of S one at a time until the reciprocal sum first reaches $3/2$ gives $Q \subseteq S$ with $\frac{3}{2} \le H_Q := \sum_{p\in Q} 1/p < 2$ (the last term is at most $1/2$) and $|Q| \le \pi(y) \le y$. For integers $r \ge 0$, $\min(r, 4) \ge r - \frac{1}{7}\binom{r}{2}$ (for $r \ge 5$ the difference is $(r-7)(r-8)/14 \ge 0$). Hence, with $E_1(Q) = \sum_{p\in Q}\lfloor m/p\rfloor$ and $E_2(Q) = \sum_{\{p,q\}\subseteq Q} \lfloor m/pq\rfloor$,

$$\sum_{k\le m} \min(\omega_Q(k), 4) \ge E_1(Q) - \tfrac{1}{7}E_2(Q) \ge mH_Q - |Q| - \tfrac{m}{14}H_Q^2 \ge m\Big(\tfrac{3}{2} - \tfrac{9}{56}\Big) - y = \tfrac{75}{56}m - y,$$

because $h \mapsto h - h^2/14$ is increasing on $[\frac{3}{2}, 2]$. As $m \geq y^3$ and $y \geq 5$, $\frac{19}{56}m \geq y+1$, so $\sum_{k\leq m} \min(\Omega_P(k), 4) \geq \sum_{k\leq m} \min(\omega_Q(k), 4) \geq m+1$. □

Proof of Theorem 12.1. If $\eta \leq 2$ apply Proposition 9.1 with the certificate of Lemma 12.3; if $\eta > 2$ with that of Lemma 12.4. (For $m < 8$ one has $y = 1$, $S = \emptyset$, $\eta = 0$, and the sparse certificate consists of the prime-power terms only.) □

Remark 12.5. (a) The threshold is $2+2$: two from $\omega_L \leq 2$, which the choice $y = m^{1/3}$ forces, and two from $(r-2)^+ \leq \binom{r}{2}/3$. Taking $y = m^{1/2}$ would give $\omega_L \leq 1$ but destroy the bound $\binom{|S|}{3} \leq E_3$. The coefficients $\frac{11}{21}, -\frac{1}{7}$ are extremal for the pair–triple ansatz: the pointwise constraint at $r = 6$ gives $3a - 1 \leq 4b$, and the value bound $\frac{11}{21} - \frac{2}{7} \cdot \frac{2}{3} = \frac{1}{3}$ is an identity, so this ansatz reaches exactly $\eta = 2$. Numerically the inequality (21) holds with threshold 2 in every instance tested (Section 8), and fails with threshold 1.

(b) Theorem 12.1 concerns 0/1 weights only. It does not by itself improve the bound on $g(n)$, because the duality of Section 8 requires the hinge inequality for all fractional weights. The obvious averaging does not transfer it: for $z_p \in [0,1]$ and $P_t := \{p : z_p \geq t\}$ one has $\int_0^1 (\Omega_{P_t}(n) - 1)^+ \, dt = w_z(n) - \max_{p|n} z_p$, which exceeds $(w_z(n) - 1)^+$ (e.g. $n = pq$ with $z_p = z_q = \frac{1}{2}$), so the averaged certificate violates (8).

(c) The anchor certificates of Lemma 9.3 cannot give an absolute threshold: for $Y = 2^{41}$, $m = (8Y)^4$ and P the primes in $(Y, 8Y]$, no choice of anchors $R \subseteq P$ (even with all positive prime-power coefficients added) reaches threshold 3, while the pair–triple certificate above handles this P at threshold 2; the computation is in the engine transcript.

13 The fractional case: reduction to a sparse core

We now return to arbitrary weights $z_p \in [0,1]$ and to the function $S(n) = \sum_p \min(z_p \, \mathrm{v}_p(n), 1)$ of Section 10; recall that the hinge inequality for S with threshold c implies it for w_z (the peel $w_z = E + S$). Write

$$\alpha_{p,j} := \min(jz_p, 1) - \min((j-1)z_p, 1) \ (j \geq 1), \qquad \text{so that} \qquad \alpha_{p,j} \geq 0, \quad \sum_j \alpha_{p,j} \leq 1, \quad S(n) = \sum_p \sum_{j\geq 1} \alpha_{p,j} [p^j \mid n$$

We call the $\alpha_{p,j}$ the *atoms* of z; the results of this section use only $\alpha_{p,j} \geq 0$ and $\sum_j \alpha_{p,j} \leq 1$. Fix m and split $S = S_0 + S_1$, where S_0 collects the atoms with $p^j \leq m/64$ and S_1 the rest, and put

$$H_{64}(m, z) := \sum_{p^j \leq m/64} \frac{\alpha_{p,j}}{p^j} \qquad \text{(the mean of } S_0 \text{ over the integers).}$$

Lemma 13.1 (large atoms)**.** *If $m \geq 4096$ then $S_1(k) \leq 1$ for every $k \leq m$. Consequently $(S(k) - 65)^+ \leq (S_0(k) - 64)^+$ for $k \leq m$, while $(S_0(b) - 1)^+ \leq (S(b) - 1)^+$ for every b.*

Proof. If $p^j, q^\ell > m/64$ with $p \neq q$ both divide $k \leq m$, then $p^j q^\ell \mid k$ and $m \geq p^j q^\ell > m^2/4096 \geq m$, a contradiction. So the atoms of S_1 active at k belong to one prime, and their sum is at most 1. □

Theorem 13.2 (dense branch)**.** *If $H_{64}(m, z) \geq \frac{17}{16}$ then, for every $x \geq 0$,*

$$\sum_{k \leq m} \big(S(k) - 65\big)^+ \leq \sum_{b \in I} \big(S(b) - 1\big)^+, \tag{22}$$

and hence the same holds for w_z in place of S.

Proof. For $m \le 4096$ the left side vanishes ($S(k) \le \omega(k) \le 12 < 65$), so let $m > 4096$. By Lemma 13.1 and the affine certificate $c_1 = -1$, $c_{p^j} = \alpha_{p,j}$ ($p^j \le m/64$), which satisfies (8) for S_0 and a fortiori for S,

$$\sum_{b\in I}(S(b)-1)^+ \;\ge\; \sum_{b\in I}(S_0(b)-1)^+ \;\ge\; \sum_{k\le m} S_0(k) - m,$$

so it suffices to show $\sum_{k\le m}\min(S_0(k),64) \ge m$, because $S_0 - (S_0-64)^+ = \min(S_0,64)$ and $(S-65)^+ \le (S_0-64)^+$. Put $h_p := \sum_{p^j\le m/64}\alpha_{p,j}/p^j \le 1/p \le \frac{1}{2}$. Adding primes one at a time until the sum first reaches $\frac{17}{16}$ gives a set Q with $\frac{17}{16} \le H_Q := \sum_{p\in Q} h_p \le \frac{25}{16}$. Let $T_Q(k) \le S_0(k)$ be the part of $S_0(k)$ coming from the primes of Q, and $U_p(k) \in [0,1]$ the contribution of a single $p \in Q$. Since $p^j \le m/64$ gives $\lfloor m/p^j\rfloor \ge \frac{63}{64}m/p^j$,

$$\sum_{k\le m} T_Q(k) = \sum_{p\in Q}\sum_{p^j\le m/64}\alpha_{p,j}\Big\lfloor\frac{m}{p^j}\Big\rfloor \ge \tfrac{63}{64}H_Q m \ge \tfrac{1071}{1024}m.$$

For the second moment, $\sum_{k\le m} U_p(k)^2 \le \sum_{k\le m} U_p(k) \le m h_p$ and, for $p \ne q$, $\sum_{k\le m} U_p(k)U_q(k) = \sum_{j,\ell}\alpha_{p,j}\alpha_{q,\ell}\lfloor m/p^j q^\ell\rfloor \le m h_p h_q$; hence $\sum_{k\le m} T_Q(k)^2 \le m(H_Q + H_Q^2) \le \frac{1025}{256}m$. Finally $\min(t,64) \ge t - t^2/256$ for all $t \ge 0$ (the right side is at most t, and at most 64 with maximum at $t = 128$), so

$$\sum_{k\le m}\min(S_0(k),64) \;\ge\; \sum_{k\le m}\min(T_Q(k),64) \;\ge\; \Big(\tfrac{1071}{1024} - \tfrac{1025}{65536}\Big)m = \tfrac{67519}{65536}\,m \;\ge\; m. \qquad \square$$

Corollary 13.3 (reduction to a sparse core)**.** *Suppose that for every $m > 4096$ and every system of atoms $\alpha_{p,j} \ge 0$ with $\sum_j \alpha_{p,j} \le 1$, supported on $p^j \le m/64$ and with $\sum \alpha_{p,j}/p^j < \frac{17}{16}$, the function $S_0 = \sum \alpha_{p,j}[p^j \mid \cdot]$ satisfies*

$$\sum_{k\le m}\big(S_0(k)-64\big)^+ \le \sum_{b\in I}\big(S_0(b)-1\big)^+ \qquad \textit{for every } x \ge 0. \tag{23}$$

Then (22) *holds for all weights, all m and all x, and consequently $g(n) \le 81\,n$ for all n.*

Proof. For $m \le 4096$ the left side of (22) is 0; for $m > 4096$ and $H_{64} \ge \frac{17}{16}$ apply Theorem 13.2; otherwise (23) and Lemma 13.1 give (22). The bound on $g(n)$ follows as in Corollary 10.2: the proof of Theorem 8.2 with threshold 65 in place of 2 gives $g(n) \le (65+16)n$. $\square$

Remark 13.4. (a) The constants are one point on a curve: peeling at m/R costs one unit of threshold and needs $m \ge R^2$, and the same computation closes the dense branch for $H_R \ge H_{\text{lo}}$ whenever $\frac{R-1}{R}H_{\text{lo}} - \big(H_{\text{lo}} + \frac{1}{2} + (H_{\text{lo}} + \frac{1}{2})^2\big)/(4R) > 1$; for $R = 64$ this allows $H_{\text{lo}} = \frac{133}{128}$.

(b) The sparse core (23) is the hard half of the problem, not the easy one: it is exactly the regime in which the affine certificate fails (Corollary 9.2). Its left side vanishes unless some $k \le m$ has $S_0(k) > 64$, which requires $\omega(k) > 64$, so (23) is vacuous for m below the product of the first 65 primes ($6.108 \cdot 10^{127}$); it cannot be tested numerically at this threshold, and any finite check must be run at a scaled-down threshold.

(c) Three natural ways of transferring Theorem 12.1 to fractional weights fail. Positive combinations of the 0/1 inequalities for the level sets $\{p : z_p \ge t\}$ do not control the right side of (22), because the level-set hinges $\sum_t(\Omega_{P_t} - 1)^+$ can be positive where $(S-1)^+ = 0$. The abstract principle "an inequality between positive combinations of hinges that holds at all 0/1 vertices holds on $[0,1]^P$" is false: with nine coordinates, seven copies of the full set on the left and the 36 pairs on the right, $7(t-4)^+ \le \binom{t}{2}$ holds for every support size t, but $z \equiv \frac{1}{2}$ gives $\frac{7}{2} > 0$. And no certificate supported on sets of a bounded number of atoms can dominate $(S-C)^+$ pointwise for all weights: in the equal-weight regime such a certificate is a polynomial of bounded degree in the number of active atoms, which cannot vanish on an initial segment and grow linearly afterwards. These are obstructions to particular arguments, not evidence against (23), which holds in every instance we could compute at scaled thresholds.

(d) One nonnegative certificate is available in general: order the atoms, lay their weights end to end on $[0, Z)$, and let c_A be the measure of the set of t for which the unit window $[t, t+1]$ meets exactly the atoms of A; then $c \ge 0$, c satisfies (8) for S (the divisor sum at n equals $\sum_i(\ell_i - 1)^+$ over the maximal runs ℓ_i of active atoms, which is at most $(S(n)-1)^+$), and its value is $\sum_{k\le m}\sum_i(\ell_i(k)-1)^+$ with no penalty terms. Whether some ordering makes this value exceed $\sum_{k\le m}(S_0(k)-64)^+$ on the sparse core is the question left open here.

What the two-sided counts alone cannot do

The certificate principle of Proposition 9.1 uses the window I only through $|I| = m$ and the two-sided counts $\lfloor m/d\rfloor \le N_I(d) \le \lfloor m/d\rfloor + 1$. The following example shows that these constraints, taken by themselves and without the arithmetic values of the base counts $\lfloor m/d\rfloor$, cannot yield an absolute threshold, even for 0/1 weights.

Proposition 13.5 (abstract two-sided counting is insufficient)**.** *Let q be a prime power, let Π be a projective plane of order q with point set X, $|X| = v = q^2+q+1$, and put $m = qv+1$. Let μ be the multiset of m subsets of X consisting of the v lines and $(q-1)v+1$ empty sets, and ν the multiset of m subsets consisting of each singleton q times and the full set X once. For every nonempty $A \subseteq X$ write $N_\mu(A)$, $N_\nu(A)$ for the number of members containing A. Then $0 \le N_\nu(A) - N_\mu(A) \le 1$ for every nonempty A, while*

$$\sum_{B\in\mu}\big(|B|-C\big)^+ - \sum_{B\in\nu}\big(|B|-1\big)^+ = v(q-C)+1 > 0 \qquad \textit{for every } C \le q,$$

and the mean $\sum_{B\in\mu}|B|/m = v(q+1)/(qv+1)$ is below $1+1/q$.

Proof. A singleton lies on $q+1$ lines and is counted $q+1$ times in ν; two points lie on exactly one line and are contained only in X; a set of at least three points lies on at most one line and is contained only in X. Every line has $q+1$ points, so $\sum_\mu(|B|-C)^+ = v(q+1-C)$, and $\sum_\nu(|B|-1)^+ = v-1$. $\square$

Here μ plays the role of $\{1,\dots,m\}$ (with $N_\mu(A)$ in place of $\lfloor m/d_A\rfloor$, d_A the least common multiple of the moduli in A) and ν that of the window. Consequently no proof of the hinge inequality with an absolute threshold can rely solely on $|I| = m$, the two-sided window counts for all divisors, and the feasibility of a certificate: the arithmetic values of the base table $N_K(d) = \lfloor m/d\rfloor$ must enter. They do enter already at first order: at most $\lfloor m/(q+1)\rfloor - \lfloor m/(q+2)\rfloor = q-1$ moduli d have $\lfloor m/d\rfloor = q+1$, whereas the configuration above needs q^2+q+1 atoms with that count, so it is not realised by any prime powers. (For $q = 5$ the abstract configuration beats the threshold 4 of Theorem 12.1, whose proof is a counting certificate; the theorem holds because its certificate is evaluated on the true table.) Also, the hinge sum is not monotone under $N_I(d) \ge N_K(d)$: for $m = 11021$, $P = \{101, 103, 107\}$ and I centred at $101 \cdot 103 \cdot 107$ one has

$\sum_K(\omega_P-1)^+=3$ but $\sum_I(\omega_P-1)^+=2$, although every $N_I(d)\ge\lfloor m/d\rfloor$. So the sparse core cannot be handled by a monotonicity argument at small harmonic mass either.

Two quantitative facts about the sparse core are worth recording. First, since $S_0\le\omega$, the left side of (23) vanishes unless some $k\le m$ has more than 64 distinct prime factors, so (23) is vacuous for m below the product of the first 65 primes, $6.108\cdot 10^{127}$; in particular the numerical evidence of Section 8 says nothing about it. Second, the left side is always tiny: with $a_p(k)=\sum_{p^j\le m/64}\alpha_{p,j}[p^j\mid k]\in[0,1]$ and $h_p=\sum_{p^j\le m/64}\alpha_{p,j}/p^j$ one has $\sum_{k\le m}e_\ell\big((a_p(k))_p\big)\le m\,e_\ell\big((h_p)_p\big)\le mH_{64}^\ell/\ell!$ for the elementary symmetric functions e_ℓ (distinct primes have coprime moduli), and $(t-C)^+\le e_{C+1}$ of the summands of t when they lie in $[0,1]$, so

$$\sum_{k\le m}\big(S_0(k)-64\big)^+\le\frac{mH_{64}^{65}}{65!}<6.24\cdot 10^{-90}\,m\qquad\text{when } H_{64}<\tfrac{17}{16},$$

while a single $k\le m$ with $S_0(k)>64$ forces $\sum_I(S_0-1)^+>63$, because the window contains a multiple b of k and $S_0(b)\ge S_0(k)$. A counterexample to (23) would therefore be a window of length $m\ge 6.1\cdot 10^{127}$ that is deficient, simultaneously for astronomically many moduli, within the ±1 freedom of the counts; we have not been able to rule this out, and the question is left open here.

The sparse core up to 10^{2887}

The arithmetic input that the two-sided counts lack is supplied, in a range of m that is finite but enormous, by the multiples of a single integer with many small prime factors.

Lemma 13.6 (packing)**.** *Let $x_1,\dots,x_N\in(0,1]$ with $\sum_i x_i>64$. Then there are pairwise disjoint index sets $G_1,\dots,G_{32}$ with $\frac43<\sum_{i\in G_r}x_i\le 2$ for every r.*

Proof. Call x_i large if $x_i>\frac23$. Pair the large items two by two; each pair has mass in $(\frac43,2]$. Then form further groups greedily from the remaining items (at most one large one, and items of mass at most $\frac23$): add items until the mass exceeds $\frac43$; each such group has mass at most $\frac43+\frac23=2$, and at most $1+\frac23$ if it starts with the odd large item. Every completed group has mass in $(\frac43,2]$, the leftover has mass at most $\frac43$, so the completed groups have total mass more than $64-\frac43>31\cdot 2$, and there are at least 32 of them. ☐

Theorem 13.7 (arithmetic shadows)**.** *Let $m>4096$ and let S_0 be as above. If $S_0(k)>64$ for some $k\le m$, then for every window I of m consecutive integers*

$$\sum_{b\in I}\big(S_0(b)-1\big)^+\ \ge\ \tfrac{32}{3}\big(1-6^{-31}\big)\frac{m}{k^{1/32}}\ \ge\ \tfrac{32}{3}\big(1-6^{-31}\big)\,m^{31/32}.$$

Proof. Write $a_p(k)=\sum_{p^j\le m/64,\ p^j\mid k}\alpha_{p,j}\in[0,1]$, so $S_0(k)=\sum_p a_p(k)$. Apply Lemma 13.6 to the numbers $a_p(k)>0$: this gives disjoint sets of primes $G_1,\dots,G_{32}$ with $\sum_{p\in G_r}a_p(k)>\frac43$. For each prime p let q_p be the largest power $p^j\le m/64$ with $\alpha_{p,j}>0$ and $p^j\mid k$, and put $D_r:=\prod_{p\in G_r}q_p$. The D_r are pairwise coprime, $\prod_r D_r\mid k$, and each $D_r\ge 6$ (a group contains at least two primes, since one prime carries mass at most 1). If $D_r\mid n$ then every atom of a prime of G_r that is active at k is active at n, so $a_p(n)\ge a_p(k)$ for $p\in G_r$; hence if exactly $t\ge1$ of the D_r divide n then

$S_0(n) > \frac{4t}{3}$ and $(S_0(n)-1)^+ > \frac{4t}{3} - 1 \geq \frac{t}{3}$. Thus the nonnegative certificate $c_{D_r} = \frac{1}{3}$ satisfies (8) for S_0, and by Proposition 9.1

$$\sum_{b \in I} (S_0(b) - 1)^+ \;\geq\; \tfrac{1}{3} \sum_{r=1}^{32} \Big\lfloor \frac{m}{D_r} \Big\rfloor \;\geq\; \tfrac{1}{3}\big(1 - 6^{-31}\big) \sum_{r=1}^{32} \frac{m}{D_r},$$

because $m/D_r \geq \prod_{s \neq r} D_s \geq 6^{31}$. Finally $\sum_r 1/D_r \geq 32\big(\prod_r D_r\big)^{-1/32} \geq 32\, k^{-1/32}$ by the arithmetic–geometric mean inequality. ☐

Corollary 13.8. *The sparse-core inequality* (23) *holds for every atom system and every window whenever* $4096 < m \leq 10^{2887}$*. Consequently the hinge inequality with threshold* 65 *holds for all weights and all* $m \leq 10^{2887}$*, and*

$$g(n) \leq 81\, n \qquad \text{for all } n \leq 10^{962}.$$

Proof. If the left side of (23) is positive, some $k \leq m$ has $S_0(k) > 64$, and Theorem 13.7 together with the moment bound above gives $R \geq \frac{32}{3}(1 - 6^{-31}) m^{31/32}$ and $L < 6.24 \cdot 10^{-90} m$; so $L \leq R$ as soon as $m^{1/32} \leq \frac{32}{3}(1 - 6^{-31})/(6.24 \cdot 10^{-90})$, which holds for $m \leq 10^{2887}$. The hinge inequality follows from Theorem 13.2 and Lemma 13.1, and the bound on $g(n)$ from the proof of Theorem 8.2 with threshold 65, since $a_n < 8n^3 \leq 10^{2887}$ for $n \leq 10^{962}$ (and $2n$ integers suffice when $a_n \geq 8n^3$). ☐

Remark 13.9. The bound $g(n) \leq 81n$ is therefore linear for every n that can be written down, and $g(n) \leq (16 + 20 \ln\ln(8n^3))n$ beyond. What is missing for all n is a version of Theorem 13.7 that charges every integer $k \leq m$ with $S_0(k) > 64$ rather than one: the shadows of different k may coincide, and the natural way of charging each k to its own multiples in the window fails by an unbounded factor on suitable windows defined by congruence conditions (engine transcript, round 12). The threshold could be raised instead: with C in place of 64 the same argument gives $R \geq c\, m^{1-2/C}$ against $L \leq mH^{C+1}/(C+1)!$, so the hinge inequality holds with a threshold of order $\sqrt{\ln m / \ln\ln m}$ for all m, which is weaker than Theorem 11.1 for large m.

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